WESTSÄCHSISCHE HOCHSCHULE ZWICKAU

FACULTY OF ELECTRICAL ENGINEERING

MASTER OF SCIENCE IN EMBEDDED SYSTEMS AND AUTONOMOUS SYSTEMS

Interim Project Documentation

24V / 250W BRUSHED DC MOTOR DRIVER

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Course: ELT10140 – Practical Course in Electric Drive Systems

Abstract

Contents

List of Figures

List of Tables

Chapter 1

Introduction

Electric drive systems represent a core area of modern electrical engineering in which power electronics, control theory and embedded software converge to convert electrical energy into controlled mechanical motion. The design of such systems requires simultaneous consideration of multiple engineering disciplines, including semiconductor device physics, thermal management, feedback control, signal conditioning and real-time software architecture. This chapter introduces the motivation of the project, defines the problem scope and boundary conditions, and reviews the relevant technical background on DC motor drives and microcontroller-based motor control.

1.1 Motivation

The control of electric machines has been a central topic in electrical engineering since the introduction of thyristor-based DC drives in the 1960s [1]. Although brushless and AC drive technologies have largely replaced brushed DC motors in high-performance industrial applications, brushed DC motors remain widely used in cost-sensitive applications such as small electric vehicles, actuators and laboratory equipment. Their simple construction – a permanent-magnet stator and a wound armature with mechanical commutator – results in a straightforward relationship between applied terminal voltage and shaft speed, making them a suitable platform for studying fundamental drive concepts without the added complexity of field-oriented control or space-vector modulation required by AC machines. From a control-theoretic perspective, the brushed DC motor is described by a second-order linear system coupling the electrical circuit dynamics to the mechanical load dynamics. The electrical subsystem is governed by Kirchhoff's voltage law applied to the armature circuit:

$$V_a = R_a\, i_a + L_a \frac{di_a}{dt} + K_e\, \omega \tag{1.1}$$

where V_a is the applied armature voltage, R_a and L_a are the armature resistance and inductance, i_a is the armature current, K_e is the back-EMF constant and ω is the angular velocity. The mechanical subsystem is described by Newton's second law for rotational motion:

$$J \frac{d\omega}{dt} = K_t\, i_a - B\, \omega - T_L \tag{1.2}$$

where J is the moment of inertia, K_t is the torque constant, B is the viscous friction coefficient and T_L is the load torque. These two coupled differential equations form the plant model that any speed or current controller must regulate. Pulse-width modulation provides an efficient mechanism to control the average voltage applied to the motor without the power dissipation penalties of linear regulation. In a PWM-based drive, a power transistor switches the full supply voltage across the motor at a fixed frequency f_{PWM}, and the average voltage is set by the duty cycle D:

$$\bar{V}_a = D \cdot V_{supply} \tag{1.3}$$

When the PWM frequency is chosen well above the electrical time constant of the motor ($\tau_e = L_a/R_a$), the resulting current ripple is small and the motor responds to the average voltage as if a continuous variable supply were applied [1]. The practical implementation of a PWM motor drive, however, involves considerably more than the control loop itself. The power stage must switch currents of several amperes with low conduction and switching losses. The gate driver must translate the microcontroller logic levels to the voltage levels required by the power transistors, often using a bootstrap technique for the high-side device. Current must be measured in real time for overcurrent protection and, optionally, for closed-loop current control. Finally, the complete system must be implemented on a printed circuit board with appropriate trace sizing, decoupling, thermal management and mechanical integration. This project addresses all of these aspects in the context of a 24 V, 250 W brushed DC motor drive, providing practical experience in the full development cycle from specification through design, assembly, testing and documentation.

1.2 Problem Statement

The objective of this project is to design, build and verify an electronic control system for a Unite Motor MY1016 brushed DC motor rated at 24 V and 250 W. The system must provide adjustable speed control through a user interface consisting of push-buttons and indicator LEDs, and must protect itself and the motor against overcurrent, motor blockage and other fault conditions without damage to any component. The main electronic subsystems – microcontroller (ATmega328P on an Arduino Nano module), half-bridge gate driver (IR2104), power MOSFETs (IRFZ44N), current sensing (shunt resistor with LM358 amplifier), voltage regulation (L7812 linear and LM2596 buck converter) and input protection (TVS diode and fuse) – are integrated on a single two-layer PCB. The motor and the 24 V laboratory power supply are connected as external components.The central engineering challenge is to ensure that all subsystems interact safely and reliably at the electrical, thermal and temporal interfaces. Specifically:

- The gate driver must provide sufficient gate-source voltage to fully enhance the power MOSFETs, including the high-side device whose source terminal floats at the switching-node potential.
- The current measurement chain must have sufficient gain and bandwidth to detect an overcurrent condition within one PWM cycle, while rejecting switching noise.
- The PCB layout must carry the full motor current ($\geq$ 13.7 A rated) with acceptable resistive losses and temperature rise, in accordance with IPC-2221.
- The firmware must implement a deterministic, non-blocking control loop with defined state transitions, soft-start, stall detection and fail-safe shutdown.

The scope of this work is limited to a laboratory prototype assembled on a wooden baseplate in an open-frame configuration. Direction reversal and regenerative braking are not required. The project follows the complete development cycle defined by the course: specification, design, procurement, assembly, test, documentation and presentation.

1.3 State of the Art

The control of brushed DC motors using power electronic converters has been a well-established practice in electrical engineering for several decades. Early DC drive systems relied on thyristor-based phase-controlled rectifiers to regulate the average voltage applied to the motor armature from an AC supply. With the widespread availability of power MOSFETs and insulated-gate bipolar transistors in the 1980s and 1990s, chopper-based DC drives using pulse-width modulation became the dominant topology for battery-supplied and low-voltage DC motor applications. These PWM-based drives offer faster dynamic response, lower harmonic distortion and simpler control compared to phase-controlled rectifiers, and they remain the standard approach for brushed DC motor control in cost-sensitive applications such as small electric vehicles, industrial actuators and laboratory equipment [1, 2]. The power stage of a PWM motor drive is typically implemented as either a single low-side switch with a freewheeling diode, a half-bridge with two active switches, or a full H-bridge with four switches. The single-switch topology is the simplest but limits operation to a single quadrant, meaning only unidirectional speed control is possible. The half-bridge extends this by placing a high-side and a low-side MOSFET in a totem-pole configuration, where the midpoint (switching node) connects to one terminal of the motor. In synchronous rectification mode, the low-side MOSFET conducts during the off-time instead of a passive diode, which reduces conduction losses because the MOSFET channel resistance is typically much lower than the forward voltage drop of even a Schottky diode. The full H-bridge enables four-quadrant operation, including forward and reverse motoring as well as regenerative braking, but requires twice the number of switches and a more complex gate drive scheme. Since direction reversal is not required in many practical applications, the half-bridge remains a common and cost-effective choice [1]. A central challenge in half-bridge and H-bridge topologies is the gate drive of the high-side N-channel MOSFET, whose source terminal is connected to the switching node and therefore floats between ground and the supply voltage during PWM operation. To turn the high-side device fully on, a gate-source voltage of at least 10 V is required, referenced to this floating source potential. The bootstrap technique provides a simple and widely used solution to this problem. A small capacitor is charged through a diode from a fixed voltage rail during the interval when the low-side switch is on and the switching node is near ground. When the high-side switch turns on and the switching node rises, the charged capacitor floats with it and provides the gate-drive energy. Dedicated integrated gate driver ICs, such as the IR2104 from Infineon Technologies, combine the bootstrap supply management, level-shifting logic, internal dead-time generation and shutdown functionality in a single package. The application notes accompanying these devices provide detailed guidelines for bootstrap capacitor sizing, diode selection and layout recommendations [3, 4]. Current measurement is an essential function in any motor drive that must provide overcurrent protection or closed-loop current control. In low-voltage drives, the most common approach is to insert a low-value resistive shunt (typically 1–100 mΩ) in the motor current path and to amplify the resulting voltage using an operational amplifier or a dedicated current-sense amplifier. The amplified signal is then sampled by the microcontroller's analog-to-digital converter. The main advantages of this approach are low cost, simplicity and direct proportionality between the shunt voltage and the instantaneous current. The main disadvantages are the additional power dissipation in the shunt ($P = I^2R$) and the sensitivity to noise coupling from the switching node, which requires careful PCB layout with Kelvin-connected sense traces [5]. Alternative current sensing methods include Hall-effect current sensors, which provide galvanic isolation but at higher cost and typically lower bandwidth, and current-sense MOSFETs with integrated mirror elements, which are less common in discrete designs. The control and protection functions of a modern motor drive are typically implemented on a low-cost microcontroller. Devices such as the Atmel (now Microchip)

ATmega328P provide hardware timer/counter modules capable of generating PWM signals at frequencies of 20 kHz or higher with duty-cycle resolution of 10 bits or more, without continuous CPU intervention. The same device provides a successive-approximation ADC for current and voltage measurements, general-purpose I/O for user interface elements (buttons, LEDs, displays) and serial communication interfaces for diagnostics. The firmware implements a state machine that manages the operating modes of the drive, including start-up sequencing with current-limited soft-start ramps, steady-state speed control with discrete or continuous duty-cycle adjustment, fault detection (overcurrent, stall, over-temperature) and safe shutdown with defined output states. The non-blocking programming style, in which the main loop polls timing counters and sensor readings without using blocking delay functions, ensures that the protection logic remains responsive at all times [6]. Thermal management is a further consideration in the design of compact motor drives. The power dissipation in the MOSFETs, freewheeling diode and shunt resistor must be conducted away from the semiconductor junctions to the ambient environment through a thermal path that includes the component package, any thermal interface material, the PCB copper and, optionally, an external heatsink. The steady-state junction temperature is estimated using the thermal resistance chain from junction to ambient ($R_{\theta,JA}$) and must remain below the maximum rating specified in the device datasheet. For surface-mount packages such as D2PAK, the thermal resistance is strongly influenced by the copper area on the PCB, making the layout a critical factor in the thermal design [7]. The present project applies these established techniques in a practical laboratory prototype. A half-bridge power stage with IRFZ44N MOSFETs is driven by an IR2104 gate driver with a BAS16J bootstrap diode. Current is sensed using a 10 mΩ shunt resistor amplified by an LM358 operational amplifier. The control, protection and user interface functions are implemented on an ATmega328P microcontroller (Arduino Nano module). All subsystems are integrated on a single two-layer PCB, while the motor and 24 V power supply are connected as external components.

Chapter 2

Circuit Design

This chapter presents the design of the motor driver circuit from the schematic level to the printed circuit board implementation. The circuit design is structured into three main parts. First, the electrical layout is described, including the power stage, driver electronics, voltage regulation, protection elements, and measurement interfaces. Based on this, a basic thermal analysis is performed to estimate the main sources of power dissipation and to assess the thermal feasibility of the selected components. Finally, the PCB design is evaluated with respect to component placement, routing of high-current paths, thermal management, and the integration of the circuit into the overall system. The complete circuit schematic is shown in Figure 2.1; a full-page version is included in Appendix A.2.

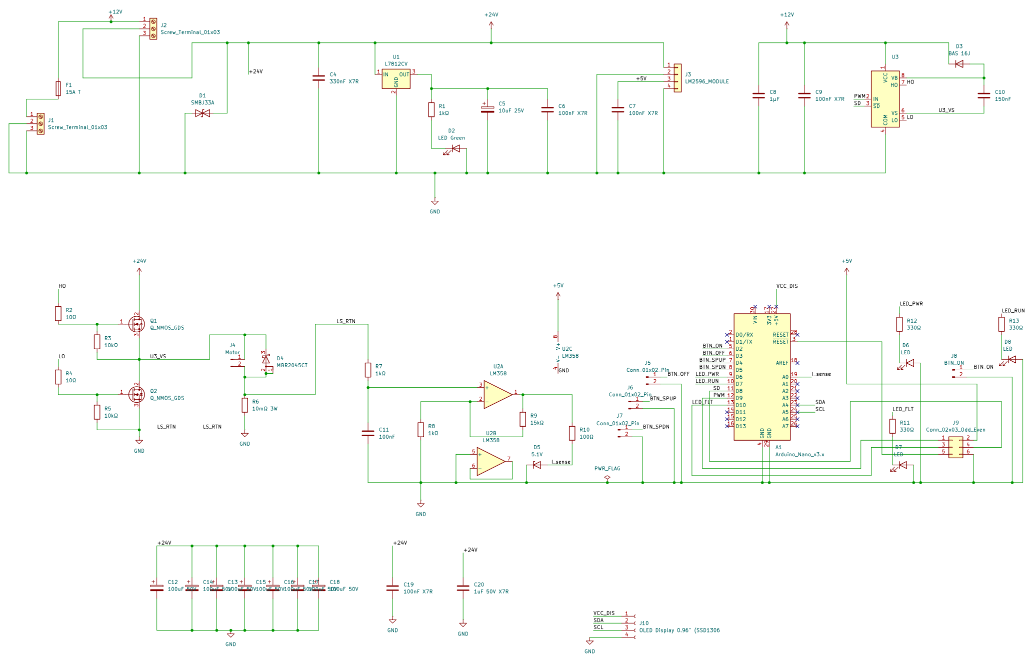


Figure 2.1: Complete schematic of the 24 V / 250 W brushed DC motor driver (KiCad export).

2.1 Electrical Layout

Compensation off Ripple Current and Ripple Voltage The input voltage of the motor driver is provided by a 24 V supply. Since the motor current is switched by the half-bridge using PWM, the DC link is subjected to pulsed current demands. These current pulses lead to voltage ripple on the supply rail due to the finite impedance of the supply path, the equivalent series resistance of the capacitors, and the parasitic inductances of the PCB traces and cables. In addition, the buck converter used for the auxiliary electronics introduces high-frequency switching components. For this reason, the 24 V rail is supported by a combination of electrolytic bulk capacitors and ceramic decoupling capacitors. The electrolytic capacitors mainly act as local energy storage for the power stage [1, 2]. In the present design, eight electrolytic capacitors are connected in parallel. Each capacitor has an equivalent series resistance of approximately

$$ESR_{\text{elko}} = 300\,\text{m}\Omega = 0.3\,\Omega. \tag{2.1}$$

Assuming an ideal current distribution between the capacitors, the equivalent series resistance of the capacitor bank is given by

$$ESR_{\text{eq}} = \frac{ESR_{\text{elko}}}{n} = \frac{0.3\,\Omega}{8} = 0.0375\,\Omega. \tag{2.2}$$

Thus, the parallel connection reduces the effective equivalent series resistance to

$$ESR_{\text{eq}} = 37.5\,\text{m}\Omega. \tag{2.3}$$

The total capacitance of the DC-link capacitor bank is calculated by

$$C_{\text{eq}} = n \cdot C_{\text{elko}}. \tag{2.4}$$

For eight capacitors with a capacitance of $100\,\mu$F each, the resulting total capacitance is

$$C_{\text{eq}} = 8 \cdot 100\,\mu\text{F} = 800\,\mu\text{F}. \tag{2.5}$$

The voltage ripple at the DC link consists of an ESR-related component and a capacitive component. The ESR-related voltage ripple can be approximated by [2]

$$\Delta V_{\text{ESR}} \approx I_{\text{ripple,rms}} \cdot ESR_{\text{eq}}. \tag{2.6}$$

For a basic estimation, the RMS ripple current of the capacitor bank can be approximated by

$$I_{\text{ripple,rms}} \approx I_{\text{load}}\sqrt{D(1-D)}. \tag{2.7}$$

The maximum value occurs at a duty cycle of approximately $D = 0.5$. With a nominal motor current of $I_{\text{load}} = 10.24\,\text{A}$, this gives

$$I_{\text{ripple,rms}} \approx 10.24\,\text{A}\sqrt{0.5(1-0.5)} = 5.12\,\text{A}. \tag{2.8}$$

Using this value, the ESR-related voltage ripple is estimated as

$$\Delta V_{\text{ESR}} \approx 5.12\,\text{A} \cdot 0.0375\,\Omega = 0.192\,\text{V}. \tag{2.9}$$

The corresponding power dissipation in the complete capacitor bank is given by

$$P_{\text{loss}} = I^2_{\text{ripple,rms}} \cdot ESR_{\text{eq}}. \tag{2.10}$$

This results in

$$P_{\text{loss}} = (5.12\,\text{A})^2 \cdot 0.0375\,\Omega \approx 0.98\,\text{W}. \tag{2.11}$$

Assuming ideal current sharing, the loss per electrolytic capacitor is

$$P_{\text{loss,elko}} = \frac{0.98\,\text{W}}{8} \approx 0.12\,\text{W}. \tag{2.12}$$

The capacitive part of the voltage ripple can be estimated by [1]

$$\Delta V_{\text{C}} \approx \frac{I_{\text{load}} \cdot D(1-D)}{C_{\text{eq}} \cdot f_{\text{PWM}}}. \tag{2.13}$$

With $I_{\text{load}} = 10.24\,\text{A}$, $D = 0.5$, $C_{\text{eq}} = 800\,\mu\text{F}$, and $f_{\text{PWM}} = 20\,\text{kHz}$, this gives

$$\Delta V_{\text{C}} \approx \frac{10.24\,\text{A} \cdot 0.5(1-0.5)}{800\,\mu\text{F} \cdot 20\,\text{kHz}} \approx 0.16\,\text{V}. \tag{2.14}$$

The estimation shows that both the ESR-related voltage ripple and the capacitive voltage ripple contribute significantly to the total voltage ripple of the DC link. The parallel connection of multiple electrolytic capacitors is therefore useful not only to increase the available capacitance, but also to reduce the effective equivalent series resistance and to distribute the ripple-current-related losses over several components. The ceramic capacitors complement the electrolytic capacitors at higher frequencies. Due to their lower equivalent series inductance and lower impedance at high frequencies, they provide a local path for fast switching current components. Thus, the electrolytic capacitors stabilize the 24 V bus against larger low-frequency current variations, while the ceramic capacitors suppress high-frequency voltage transients generated by the half-bridge and the buck converter.

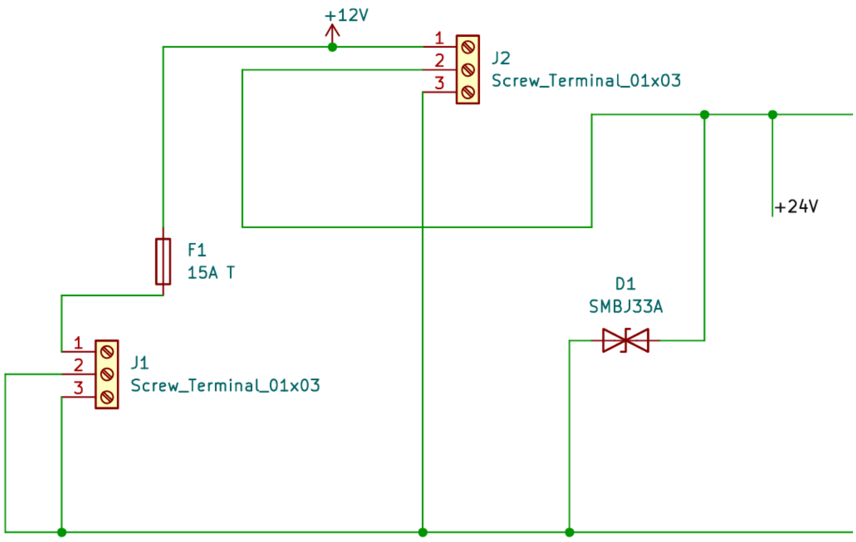


Figure 2.2: Input protection section of the schematic: screw terminals J1 and J2, 15 A fuse F1, and TVS diode D1 (SMBJ33A).

Linear Voltage Regulator The gate driver IC and the bootstrap circuit require a supply voltage in the range of 10 to 20 V. The IR2104 gate driver enforces this constraint through an internal under-voltage lockout with a positive-going threshold of $V_{\text{CCUV+}} = 8.9\,\text{V}$ (typical) and a negative-going threshold of $V_{\text{CCUV-}} = 8.2\,\text{V}$ (typical), as specified in the IR2104 datasheet [3]. Since the 5 V rail generated by the buck converter module is below this threshold, a separate 12 V supply is required. This supply is provided by the STMicroelectronics L7812CV linear voltage regulator (U1), which converts the 24 V bus to a regulated 12 V output. A 330 nF X7R ceramic capacitor (C4) is placed directly at the input of U1. This capacitor serves as both the high-frequency input bypass for the regulator and a local charge reservoir for the switching transients coupled from the power stage. The L7812 datasheet (STMicroelectronics L7800 series, Application circuits, Fig. 4) recommends a minimum input capacitance of $C_{\text{in}} \geq 0.33\,\mu\text{F}$, which is satisfied by C4. The output of U1 feeds a rail designated Net-(U1-OUT) in the netlist. Two capacitors are placed on this rail: a 10 µF / 25 V electrolytic capacitor (C5) and a 100 nF X7R ceramic capacitor (C6). The electrolytic capacitor provides bulk energy storage and improves the transient response of the regulator under sudden load steps, for example when the IR2104 charges both MOSFET gates simultaneously. The ceramic capacitor provides a low-impedance path at higher frequencies. A 25 V voltage rating for C5 provides a derating margin of more than 2× above the 12 V operating voltage. The L7812 output also supplies a green LED (D2) through a 1 kΩ series resistor (R1), which serves as a visual power-on indicator. The LED forward current is approximately

$$I_{\text{LED}} = \frac{V_{\text{out}} - V_{\text{F}}}{R_1} = \frac{12\,\text{V} - 2\,\text{V}}{1\,\text{k}\Omega} = 10\,\text{mA}. \tag{2.15}$$

The total current demand on the 12 V rail is low. The IR2104 quiescent supply current is $I_{\text{QCC}} = 150\,\mu\text{A}$ (typical) [3], and the average gate-drive current per MOSFET at 20 kHz is

$$I_{\text{gate,avg}} = Q_{\text{g}} \cdot f_{\text{PWM}} = 63\,\text{nC} \cdot 20\,\text{kHz} = 1.26\,\text{mA}. \tag{2.16}$$

Including the LED current, the total 12 V load is estimated at approximately 25 mA. The worst-case power dissipation in U1 is therefore

$$P_{\text{L7812}} = (V_{\text{in}} - V_{\text{out}}) \cdot I_{\text{out}} = (24\,\text{V} - 12\,\text{V}) \cdot 25\,\text{mA} = 0.30\,\text{W}. \tag{2.17}$$

With the TO-220 package thermal resistance of $R_{\theta,\text{JA}} = 50\,\text{K}\,\text{W}^{-1}$ (STMicroelectronics L7800 datasheet, Thermal data), the estimated junction temperature rise above ambient is $\Delta T = 0.30\,\text{W} \cdot 50\,\text{K}\,\text{W}^{-1} = 15\,\text{K}$, resulting in a junction temperature of approximately 40 °C. This is well within the maximum junction temperature of 150 °C, and no external heatsink is required.

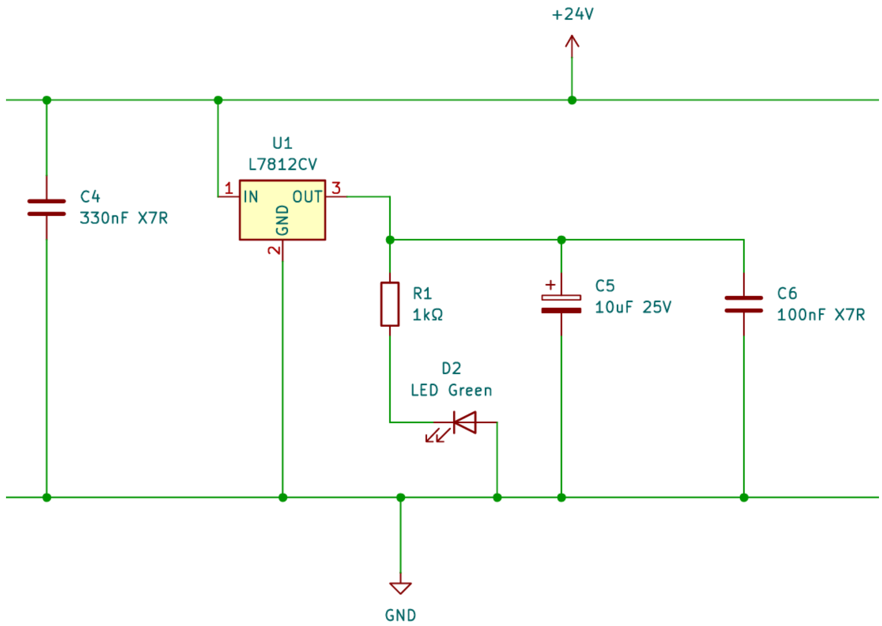


Figure 2.3: Linear voltage regulator section: L7812CV (U1) with input capacitor C4 (330 nF), output capacitors C5 (10 μF) and C6 (100 nF), power-on indicator LED D2 with series resistor R1 (1 kΩ).

Gate Driver and Bootstrap Circuit The IR2104 (U3) is a half-bridge gate driver from Infineon Technologies. It accepts a logic-level PWM input signal on pin 2 (IN) and generates non-overlapping complementary gate-drive outputs for the high-side MOSFET Q1 (via HO, pin 7) and the low-side MOSFET Q2 (via LO, pin 5). The device generates an internally fixed dead time of $t_{\mathrm{DT}} = 520\,\mathrm{ns}$ (typical, range 400 to 650 ns), which prevents simultaneous conduction of both MOSFETs [3]. An active-low shutdown input ($\overline{\mathrm{SD}}$, pin 3) allows the microcontroller to disable both outputs and force the gate driver into a safe state. The input logic thresholds are $V_{\mathrm{IH}} \geq 3.0\,\mathrm{V}$ and $V_{\mathrm{IL}} \leq 0.8\,\mathrm{V}$ [3]. Since the ATmega328P GPIO outputs swing between 0 V and 5 V [6], the logic levels are directly compatible with the IR2104 input thresholds without a level shifter. The VCC pin (pin 1) of U3 is connected to the +12 V rail. Two decoupling capacitors are placed directly on this rail at U3: a 1 µF capacitor (C8) and a 100 nF X7R ceramic capacitor (C9). These capacitors provide a local charge reservoir and a low-impedance path for the pulsed current drawn by the gate driver during switching events. The COM pin (pin 4) is connected to circuit ground. Each MOSFET gate is driven through a 10 Ω series resistor (R2 for Q1, R4 for Q2). These resistors control the gate voltage slew rate and damp parasitic oscillations in the gate loop [4]. With a 12 V gate drive supply, the peak gate current through each resistor is limited to

$$I_{\mathrm{gate,peak}} = \frac{V_{\mathrm{CC}}}{R_{\mathrm{g}}} = \frac{12\,\mathrm{V}}{10\,\Omega} = 1.2\,\mathrm{A}. \tag{2.18}$$

In practice, the peak current is limited by the output impedance of the IR2104 driver stage, which can source 210 mA and sink 360 mA [3]. The 10 Ω resistor therefore primarily serves as a damping element rather than a current limiter. A 10 kΩ pull-down resistor is placed between gate and source of each MOSFET (R3 for Q1, R5 for Q2) to ensure a defined low gate voltage during power-up, before the gate driver output stage is active. Driving the gate of the high-side N-channel MOSFET Q1 requires a gate-source voltage referenced to the source of Q1, which is the switching node. During PWM operation, the switching node floats between 0 V and approximately 24 V. A gate driver supply referenced to the fixed ground rail cannot directly provide this floating drive voltage. The bootstrap technique solves this problem by using a capacitor to store a charge referenced to the switching node [4]. The bootstrap diode D3 (BAS16J, SOD-323F package) and the bootstrap capacitor C10 (1 µF) form the bootstrap network. During the interval in which the low-side MOSFET Q2 is conducting, the switching node VS is pulled close to ground, and the bootstrap capacitor C10 charges through D3 from the 12 V supply to approximately

$$V_{\mathrm{C10}} \approx V_{\mathrm{VCC}} - V_{\mathrm{F,D3}} \approx 12\,\mathrm{V} - 0.7\,\mathrm{V} = 11.3\,\mathrm{V}. \tag{2.19}$$

When Q2 turns off and Q1 must be switched on, the VS node rises toward the 24 V bus. The bootstrap capacitor floats with the switching node, so the VB pin of the gate driver rises to $V_{\mathrm{S}} + V_{\mathrm{C10}}$, which is above the supply voltage. The IR2104 uses VB as its high-side supply and drives HO between VB and VS, providing the required gate-source voltage to turn Q1 fully on [3, 4]. During this interval, D3 is reverse-biased and prevents C10 from discharging back into the 12 V supply. The BAS16J was selected for its fast switching characteristic ($t_{\mathrm{rr}} \leq 4\,\mathrm{ns}$), which minimises charge leakage from the bootstrap capacitor during high-frequency switching. Its reverse voltage rating of 75 V provides adequate margin. The charge removed from C10 per switching cycle is dominated by the total gate charge of Q1. With $Q_{\mathrm{g}} = 63\,\mathrm{nC}$ and the IR2104 bootstrap quiescent current $I_{\mathrm{QBS}} = 55\,\mu\mathrm{A}$ (maximum) [3], the total charge drawn per cycle at the worst-case maximum on-time of $t_{\mathrm{on}} = 50\,\mu\mathrm{s}$ is

$$Q_{\mathrm{total}} = Q_{\mathrm{g}} + I_{\mathrm{QBS}} \cdot t_{\mathrm{on}} = 63\,\mathrm{nC} + 55\,\mu\mathrm{A} \cdot 50\,\mu\mathrm{s} = 65.75\,\mathrm{nC}. \tag{2.20}$$

The resulting voltage droop on the bootstrap capacitor per cycle is

$$\Delta V_{\mathrm{C10}} = \frac{Q_{\mathrm{total}}}{C_{10}} = \frac{65.75\,\mathrm{nC}}{1\,\mu\mathrm{F}} = 0.066\,\mathrm{V}. \tag{2.21}$$

This droop of approximately 66 mV is small compared to the 11.3 V bootstrap voltage, confirming that the 1 µF capacitor provides sufficient energy storage for reliable high-side gate drive at 20 kHz, consistent with the bootstrap design methodology outlined in [4]. The freewheel diode D4 (MBR2045CT) provides a current path for the motor inductive current during the dead time when both MOSFETs are off. The MBR2045CT is a dual common-cathode Schottky rectifier rated for 20 A average forward current and 45 V reverse voltage (Vishay document 87591). Its low forward voltage drop of approximately 0.84 V minimises the conduction loss during the freewheeling interval compared to the body diode of the MOSFET [1].

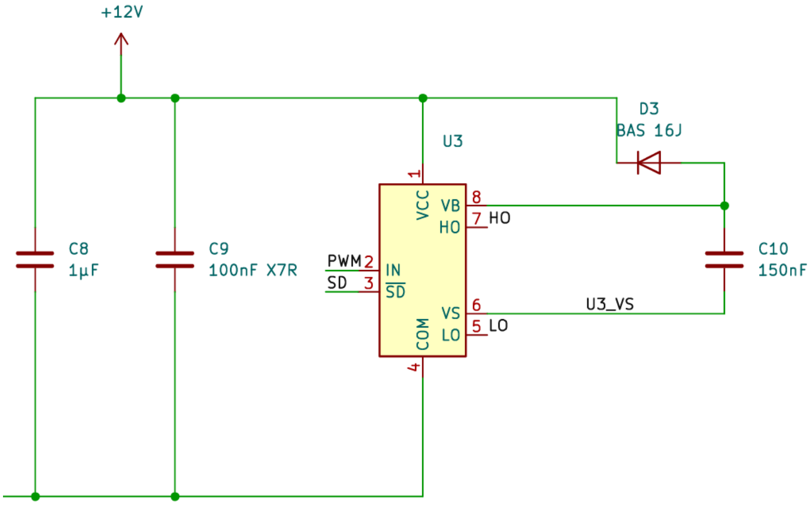


Figure 2.4: Gate driver and bootstrap circuit: IR2104 (U3) with VCC decoupling capacitors C8 (1 μF) and C9 (100 nF), bootstrap diode D3 (BAS 16J) and bootstrap capacitor C10.

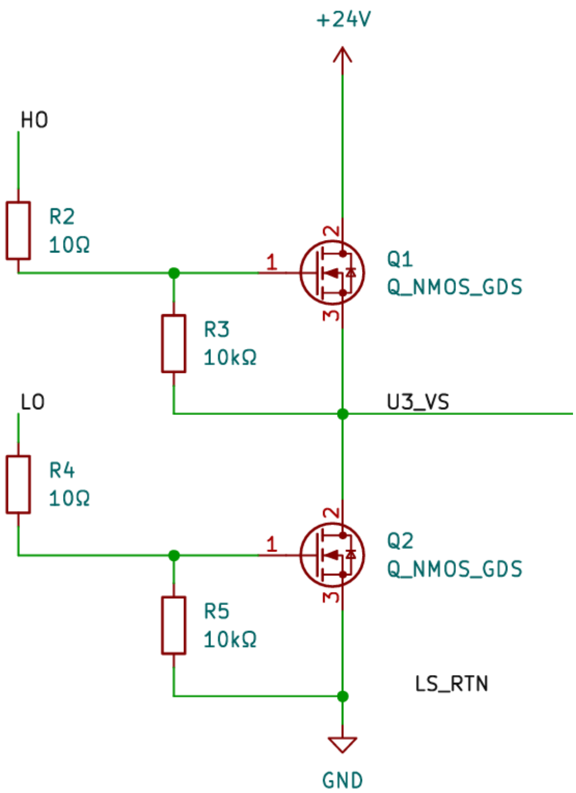


Figure 2.5: Half-bridge power stage: high-side MOSFET Q1 and low-side MOSFET Q2 (IRFZ44N) with gate resistors R2, R4 (10 Ω) and gate pull-down resistors R3, R5 (10 kΩ).

Buck Converter The 5 V supply for the Arduino Nano (A1), the LM358 current-sense amplifier (U2), the indicator LEDs (D6, D7, D8) and the button interface connectors is generated by a pre-assembled LM2596-based buck converter module connected at J3. The module input (J3 pin 1) is connected to the +24 V bus, and the output (J3 pin 3) provides the regulated +5 V rail. This connection was verified in the PCB netlist, which assigns J3 pin 1 to the +24 V net and J3 pin 3 to the +5 V net. The pre-assembled module integrates the LM2596 switching regulator IC, inductor, Schottky catch diode and feedback network, avoiding the layout-critical aspects of a discrete switching regulator design [2]. A 100 nF X7R ceramic capacitor (C7) is placed on the +5 V rail on the PCB. This capacitor provides additional local high-frequency decoupling at the PCB level and supplements the output capacitors already present on the module itself. The +5 V rail connects directly to the power supply pins of the Arduino Nano (A1), the supply pin of the LM358 operational amplifier (U2, pin 8), the OLED display connector (J10) and the indicator LEDs.

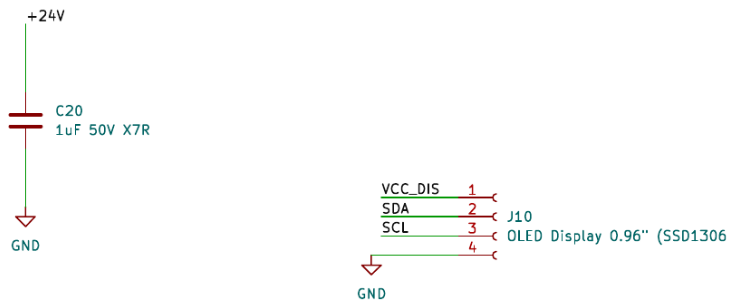


Figure 2.6: Buck converter module and OLED display connector: LM2596 module at J3 (24 V → 5 V), local decoupling capacitor C20 (1 μF) and OLED display connector J10 (SSD1306, I^2C).

The motor current is measured using a low-side shunt resistor and a non-inverting amplifier, following the standard current-sensing approach for motor drive applications [5]. A 10 mΩ / 3 W shunt resistor (R6) is placed in the LS_RTN net between the source of the low-side MOSFET Q2 and the ground return. The voltage developed across the shunt is proportional to the instantaneous motor current. At the rated motor current, this voltage is

$$V_{\text{shunt}} = I_{\text{motor}} \cdot R_{\text{shunt}} = 13.7\,\text{A} \cdot 10\,\text{m}\Omega = 0.137\,\text{V}. \tag{2.22}$$

This voltage is too small to be read directly by the microcontroller ADC with useful resolution. It is therefore amplified by the LM358 dual operational amplifier (U2). Channel A of the LM358 is configured as a non-inverting amplifier with the shunt voltage applied to the non-inverting input (pin 3) through a 1 kΩ input resistor (R7) and a 100 nF input filter capacitor (C11). The gain is set by the feedback network consisting of $R_8 = 1\,\text{k}\Omega$ (inverting input to ground) and $R_9 = 15\,\text{k}\Omega$ (feedback from output to inverting input):

$$G = 1 + \frac{R_9}{R_8} = 1 + \frac{15\,\text{k}\Omega}{1\,\text{k}\Omega} = 16. \tag{2.23}$$

At the rated motor current, the amplifier output voltage is

$$V_{\text{out,sense}} = G \cdot V_{\text{shunt}} = 16 \cdot 0.137\,\text{V} = 2.19\,\text{V}. \tag{2.24}$$

The amplifier output is connected to the Arduino ADC input A0 through a 100 Ω series resistor (R10), with a 5.1 V Zener diode (D5) providing overvoltage protection at the ADC input. The ATmega328P 10-bit ADC with $V_{\text{REF}} = 5.0\,\text{V}$ [6] produces the following count at rated current:

$$\text{ADC}_{\text{rated}} = \frac{V_{\text{out,sense}}}{V_{\text{REF}}} \cdot 1023 = \frac{2.19\,\text{V}}{5.0\,\text{V}} \cdot 1023 \approx 448. \tag{2.25}$$

At the overcurrent protection threshold of 20 A, the corresponding values are

$$V_{\text{out,20A}} = 16 \cdot (20\,\text{A} \cdot 10\,\text{m}\Omega) = 3.20\,\text{V}, \tag{2.26}$$

$$\mathrm{ADC}_{20\mathrm{A}} = \frac{3.20\,\mathrm{V}}{5.0\,\mathrm{V}} \cdot 1023 \approx 655. \tag{2.27}$$

This output voltage of 3.20 V is within the linear output range of the LM358, which can swing to approximately $V_{\mathrm{CC}} - 1.5\,\mathrm{V} = 3.5\,\mathrm{V}$ on the positive side (TI datasheet SNOSBT3). The LM358 input offset voltage of up to 7 mV is amplified to $16 \times 7\,\mathrm{mV} = 112\,\mathrm{mV}$ at the output, corresponding to a measurement error of approximately 0.7 A or 3.5 % of the 20 A threshold. This error is removed by a zero-offset calibration performed in firmware at power-on, a technique commonly recommended for shunt-based current sensing [5]. Channel B of the LM358 is unused; its inverting input (pin 6) is connected to its output (pin 7) to form a unity-gain buffer and prevent oscillation on the floating input.

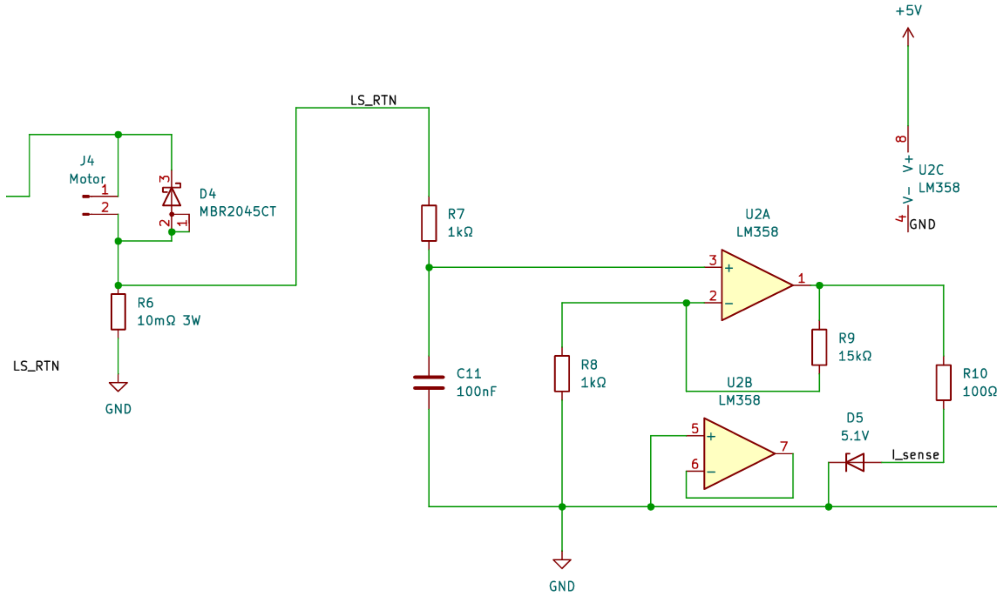


Figure 2.7: Current sensing circuit: motor output connector J4 with freewheeling diode D4 (MBR2045CT), low-side shunt resistor R6 (10 mΩ, 3 W), LM358 non-inverting amplifier (U2A, gain = 16) with input filter C11, feedback resistors R8 and R9, ADC protection resistor R10 (100 Ω) and Zener diode D5 (5.1 V). Channel B (U2B) is configured as a unity-gain buffer.

Three indicator LEDs are connected to GPIO outputs of the Arduino Nano through 330 Ω current-limiting resistors: R12 with D6 (power), R11 with D7 (fault), and R13 with D8 (run). The LED forward current at a typical forward voltage of 2 V is

$$I_{\mathrm{LED}} = \frac{V_{\mathrm{CC}} - V_{\mathrm{F}}}{R} = \frac{5.0\,\mathrm{V} - 2.0\,\mathrm{V}}{330\,\Omega} = 9.1\,\mathrm{mA}. \tag{2.28}$$

This is within the absolute maximum GPIO current of 40 mA per pin specified in the ATmega328P datasheet [6]. Four two-pin connectors (J5, J6, J7, J8) provide the push-button interface for OFF, Speed-Up, Speed-Down and ON functions. Each button connects the corresponding GPIO pin to ground when pressed; the ATmega328P internal pull-up resistors hold the pins high in the unpressed state [6]. An optional 0.96-inch OLED display (SSD1306) is connected via the I^{2}C interface (SDA, SCL) through connector J10.

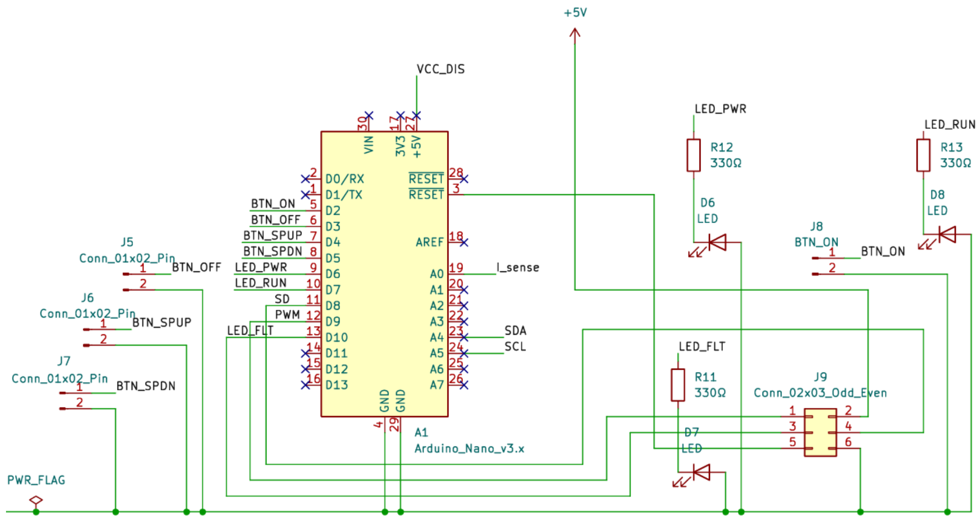


Figure 2.8: Control board and user interface: Arduino Nano (A1, ATmega328P) with button connectors J5–J8 (OFF, Speed-Up, Speed-Down, ON), ISP header J9, status LEDs D6 (power), D7 (fault) and D8 (run) with 330 Ω current-limiting resistors R11, R12 and R13.

2.2 Basic Thermal Analysis

The thermal analysis was performed in order to identify the components of the circuit that are most likely to limit the continuous operating capability of the motor driver. Since the circuit contains both high-current switching elements and low-power control electronics, not all components are equally relevant from a thermal point of view. The main focus was therefore placed on the components that either conduct the motor current directly or dissipate switching-related losses. These components are the high-side MOSFET Q1, the low-side MOSFET Q2, the freewheel diode D2, the current sense shunt R5 and the IR2104 half-bridge gate driver. The MOSFETs were considered critical because they carry the motor current during PWM operation and additionally dissipate switching losses during each commutation event [1]. The freewheel diode was included because the selected operating mode assumes diode freewheeling during the off-time of the PWM signal, which can lead to significant conduction losses. The shunt resistor was included because its losses scale quadratically with the motor current. The gate driver was also considered, although its losses are comparatively small, because it supplies the MOSFET gate charge at the PWM frequency. Other components, such as the bootstrap diode, bootstrap capacitor, operational amplifier and microcontroller, were not treated as dominant heat sources in this first-order estimate, because their expected dissipation is small compared to the high-current power stage. The analysis is based on a simplified but transparent loss model. For the nominal operating point, a supply voltage of $V_{\text{supply}} = 24\,\text{V}$, a motor current of $I_{\text{motor}} = 10\,\text{A}$, a PWM frequency of $f_{\text{PWM}} = 20,\text{kHz})$, a duty cycle of $\text{D} = 0.5$ and an ambient temperature of $T_{\text{amb}} = 25,^{\circ}\,\text{C}$ were assumed. The conduction losses of the MOSFETs were calculated from the RMS current approximation during the corresponding duty interval. For a MOSFET with drain-source on-resistance $R_{\text{DS,on}}$, the conduction loss is given by [1, 2]

$$P_{\text{cond}} = I^2 D R_{\text{DS,on}}. \tag{2.29}$$

The switching loss was estimated by assuming a linear overlap of voltage and current during the rise and fall transitions. With the drain-source voltage V_{DS}, drain current I_{D}, rise time t_{r}, fall time t_{f} and PWM frequency f_{PWM}, the switching loss is approximated as [1, 2]

$$P_{\text{sw}} = \frac{1}{2} V_{\text{DS}} I_{\text{D}} \left(t_{\text{r}} + t_{\text{f}}\right) f_{\text{PWM}}. \tag{2.30}$$

The total MOSFET loss is then obtained by adding conduction and switching losses:

$$P_{\text{MOSFET}} = P_{\text{cond}} + P_{\text{sw}}. \tag{2.31}$$

For the gate driver, the loss contribution caused by charging and discharging the MOSFET gates was estimated from the total gate charge. With the gate charge Q_g, gate voltage V_{GS} and PWM frequency f_{PWM}, the corresponding gate-drive loss is [4]

$$P_{\text{gate}} = Q_g V_{GS} f_{PWM}. \tag{2.32}$$

This loss is mainly dissipated in the gate driver and the gate resistance and is therefore not fully assigned to the MOSFET junction itself. In the diode freewheeling case, the freewheel diode was modeled using a linearized conduction loss relation. The average and RMS diode currents during the off-time are

$$I_{\text{F,av}} = I_{\text{motor}} (1 - D), \tag{2.33}$$

$$I_{\text{F,rms}} = I_{\text{motor}} \sqrt{1 - D}. \tag{2.34}$$

The diode loss was then approximated by [2]

$$P_{\text{diode}} = a I_{\text{F,av}} + b I_{\text{F,rms}}^2, \tag{2.35}$$

where (a) and (b) are the parameters of the simplified diode loss model. For the current sense resistor, the dissipated power is determined by the ohmic loss

$$P_{\text{shunt}} = I^2 R_{\text{shunt}}. \tag{2.36}$$

After calculating the individual power losses, the steady-state temperature of each component was estimated using a lumped thermal resistance model. The temperature rise above ambient is proportional to the dissipated power and the thermal resistance from the component to the ambient environment:

$$\Delta T = P_{\text{loss}} R_{\text{th}}. \tag{2.37}$$

The corresponding steady-state temperature is

$$T_{\text{ss}} = T_{\text{amb}} + P_{\text{loss}} R_{\text{th}}. \tag{2.38}$$

This model is suitable for an initial design estimate, but it does not replace detailed PCB-level thermal simulation or measurements on the final hardware. In particular, the assumed thermal resistances depend strongly on the package, copper area, airflow and heat spreading through the PCB [7]. The resulting steady-state temperatures are shown in Figure 2.9. The highest estimated temperature occurs at the high-side MOSFET Q1, which reaches approximately 108,° C under the nominal assumptions. This is mainly caused by the combination of conduction losses and switching losses. The freewheel diode D2 reaches approximately 95,° C, which makes it the second most critical component in this estimate. In contrast, the low-side MOSFET Q2 remains much cooler in the selected diode-freewheeling mode, because it only contributes switching losses and does not conduct during the off-time. The shunt resistor reaches approximately 60,° C, while the gate driver remains close to ambient temperature due to its small gate-drive loss. To estimate the transient heating behavior, each component was additionally modeled as a first-order thermal system with thermal resistance R_{th} and thermal capacitance C_{th}. The corresponding thermal time constant is

$$\tau = R_{\text{th}} C_{\text{th}}. \tag{2.39}$$

For a constant power loss, the component temperature over time is described by

$$T(t) = T_{\text{ss}} + (T_0 - T_{\text{ss}}) e^{-t/\tau}, \tag{2.40}$$

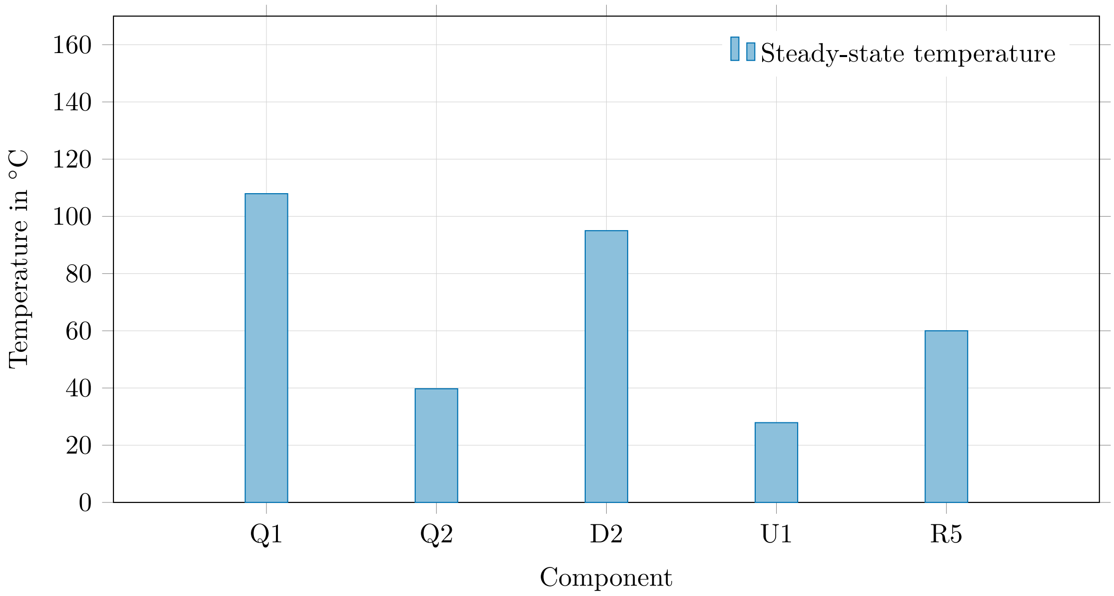


Figure 2.9: Estimated steady-state component temperatures for the nominal operating point.

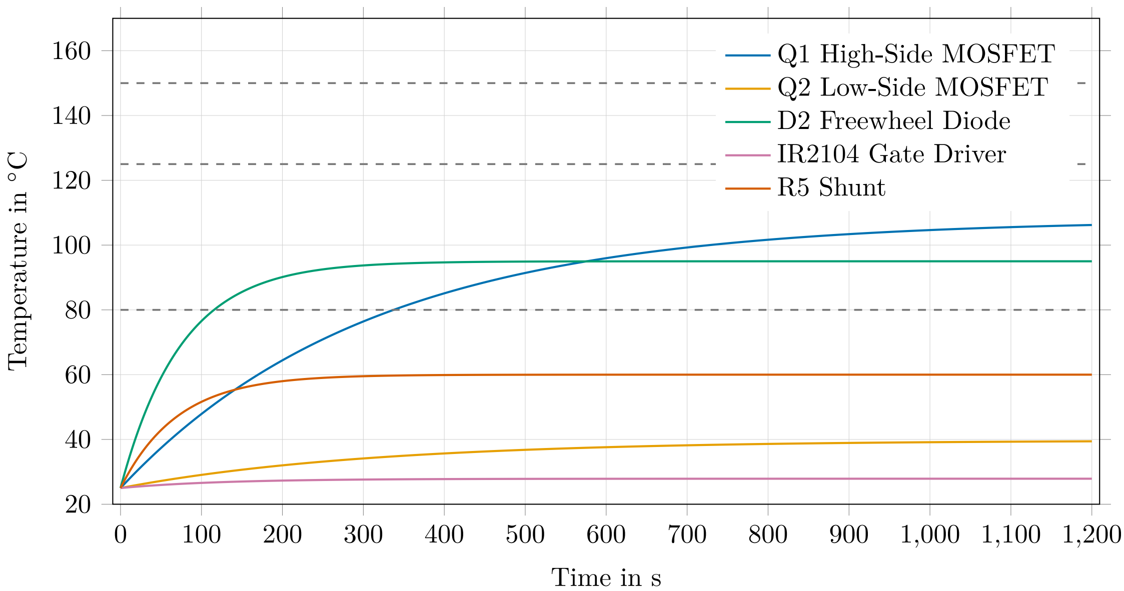


Figure 2.10: Transient temperature response using independent first-order thermal models.

where T_0 is the initial component temperature. This equation describes the exponential approach from the initial temperature to the steady-state temperature. Figure 2.10 shows the transient response of the investigated components. The curves indicate that the high-side MOSFET and the freewheel diode are not only the hottest components in steady state, but also the most relevant components during longer operation. The result also shows that the temperature rise is not instantaneous, because the assumed thermal capacitances introduce a thermal delay. Nevertheless, for continuous operation the steady-state values remain the decisive design criterion. In order to evaluate the sensitivity of the design with respect to the motor current, a current sweep was carried out. Since several loss mechanisms depend quadratically on current, especially MOSFET conduction losses and shunt losses, the motor current has a strong influence on the thermal behavior.

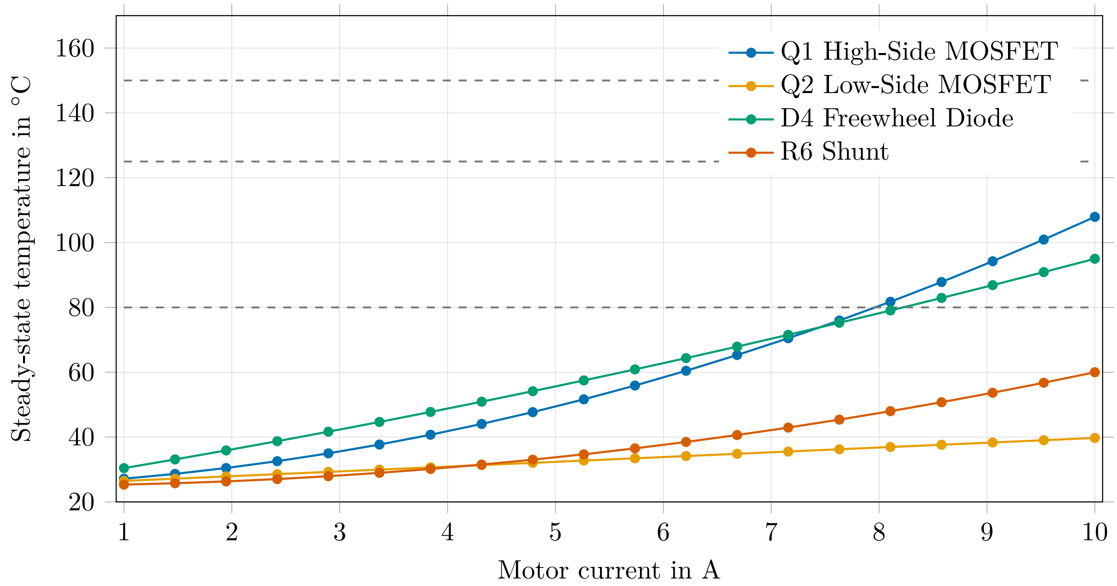


Figure 2.11: Influence of the motor current on the estimated steady-state temperature of the main power components.

As shown in Figure 2.11, the temperatures of the high-side MOSFET, freewheel diode and shunt resistor increase significantly with current. This confirms that the motor current is the dominant operating parameter for the thermal design. The freewheel diode and the high-side MOSFET become especially critical near the nominal current of 10, A, while the low-side MOSFET remains comparatively uncritical in the diode-freewheeling configuration. The PWM frequency was also varied in order to evaluate the influence of switching losses. In the simplified switching loss model, the switching loss is directly proportional to the PWM frequency [1]. Therefore, increasing the PWM frequency reduces current ripple and acoustic noise, but increases the thermal stress on the switching devices.

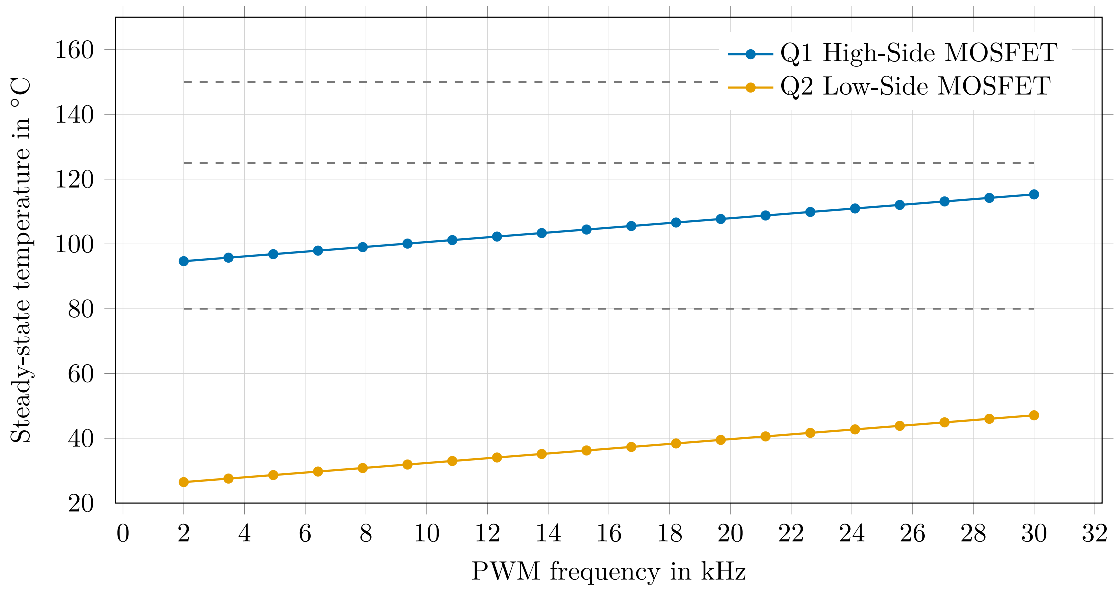


Figure 2.12: Influence of the PWM frequency on the estimated MOSFET temperature.

Figure 2.12 shows that the estimated MOSFET temperatures increase approximately linearly with the PWM frequency. The effect is more visible for Q2 because, in the selected model, Q2 only dissipates switching losses. For Q1, the switching-related increase is added to the already present conduction loss. This confirms that the selected PWM frequency should not be increased without considering the resulting thermal penalty. A further parameter sweep was performed for the duty cycle. The duty cycle changes the distribution of conduction time between the high-side MOSFET and the freewheel path. In the diode-freewheeling mode, a higher duty cycle increases the high-side MOSFET conduction interval, while reducing the diode freewheeling interval.

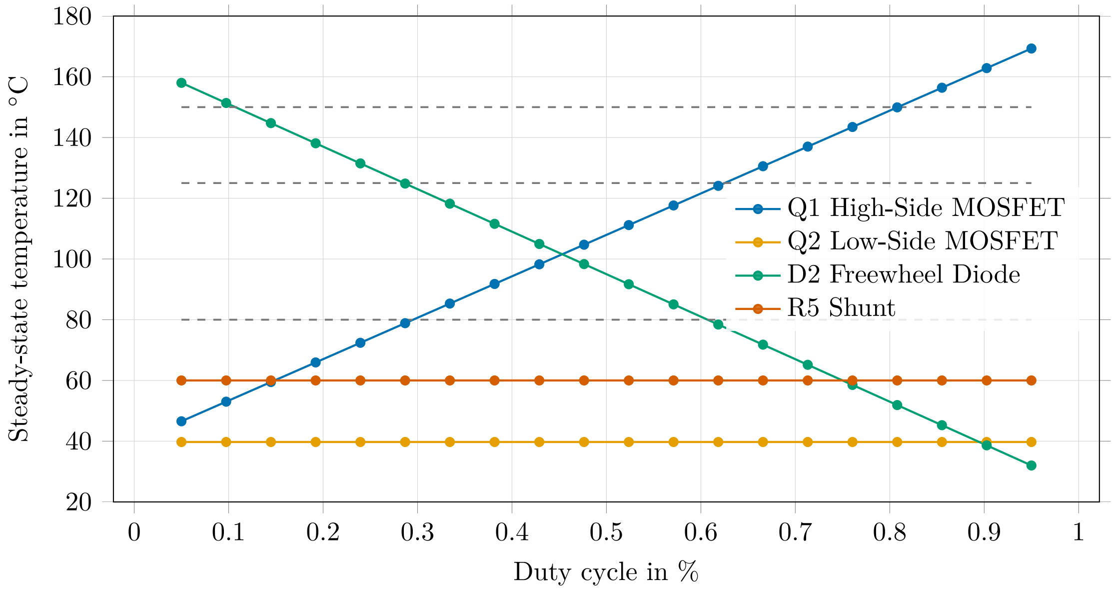


Figure 2.13: Influence of the duty cycle on the estimated steady-state temperature of the main power components.

The duty-cycle sweep in Figure 2.13 shows this thermal load shift clearly. At low duty cycles, the freewheel diode dissipates a large fraction of the power because the off-time is long. At high duty cycles, the diode loss decreases, while the high-side MOSFET temperature increases due to the longer conduction interval. The shunt temperature remains constant in this sweep because the motor current was kept constant. Because the MOSFET on-resistance increases with junction temperature, an additional feedback calculation was performed for Q1. The temperature dependence of the on-resistance was approximated by [2]

$$R_{\mathrm{DS,on}}(T) = R_{\mathrm{DS,on,25}}\left[1 + \alpha\,(T - 25\,^{\circ}\mathrm{C})\right]. \tag{2.41}$$

This feedback increases the calculated Q1 loss because a higher temperature increases $R_{\mathrm{DS,on}}$, which further increases the conduction loss. In the nominal calculation without this feedback, Q1 reaches approximately 108,° C. With the temperature-dependent on-resistance included, the estimated Q1 temperature increases to approximately 151,° C. This result shows that the simple constant $R_{\mathrm{DS,on}}$ calculation can underestimate the MOSFET temperature and that cooling measures or a MOSFET with lower on-resistance should be considered for continuous operation. Finally, a simplified PCB-coupled thermal network was evaluated to represent heat spreading through the circuit board [7]. In this model, each component is connected to a common PCB thermal node by a component-to-PCB thermal resistance. The PCB node is then connected to the ambient environment by a PCB-to-ambient thermal resistance. For each component (i), the thermal differential equation is

$$C_i \frac{\mathrm{d}T_i}{\mathrm{d}t} = P_i - \frac{T_i - T_{\mathrm{PCB}}}{R_{i,\mathrm{PCB}}}. \tag{2.42}$$

The PCB temperature is described by

$$C_{\text{PCB}} \frac{\mathrm{d}T_{\text{PCB}}}{\mathrm{d}t} = \sum_i \frac{T_i - T_{\text{PCB}}}{R_{i,\text{PCB}}} - \frac{T_{\text{PCB}} - T_{\text{amb}}}{R_{\text{PCB,amb}}}. \tag{2.43}$$

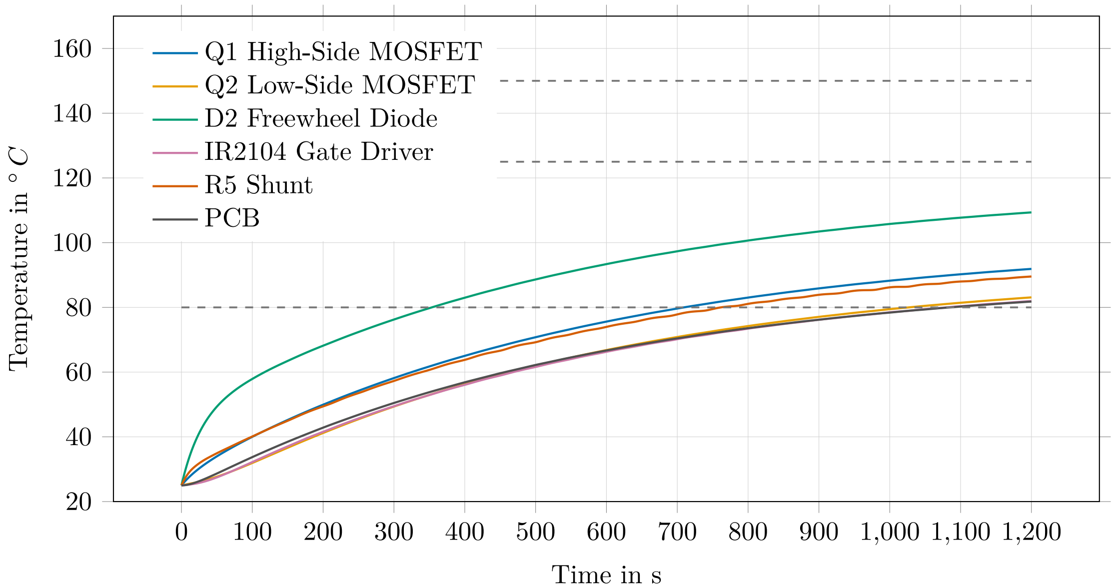


Figure 2.14: Transient thermal response of the simplified PCB-coupled thermal network.

The coupled thermal network in Figure 2.14 shows that the PCB temperature rises together with the component temperatures. This model is still a simplified approximation, but it is more realistic than treating all components as thermally independent. The result indicates that the board itself becomes a relevant thermal path and that sufficient copper area, thermal vias, component spacing and possible heatsinking should be considered in the PCB design [7]. Overall, the basic thermal analysis identifies Q1 and D2 as the most critical components under the assumed diode-freewheeling operating point. The shunt resistor is relevant but remains within an acceptable range in the nominal estimate, while the gate driver is thermally uncritical. The analysis therefore supports the need for careful MOSFET selection, diode cooling, and a PCB layout with adequate heat spreading capability.

2.3 PCB Design

The printed circuit board was designed as a two-layer PCB for the integration of the motor power stage, the voltage supply, the gate driver circuit, the current measurement and the microcontroller interface. The layout was created with the objective of separating the high-current power path from the low-power control and signal circuitry as far as possible, while still keeping the switching loop of the half-bridge compact [4]. Figure 2.15 shows the PCB layout with a board size of 130 mm × 93 mm. The input supply and bulk capacitors are placed along the upper edge of the PCB. From there, the 24 V power rail is routed with wide copper areas towards the fuse, the half-bridge stage and the motor output. The control electronics, including the Arduino Nano interface and the low-power signal routing, are located mainly in the lower and central part of the board. This placement reduces unnecessary overlap between high-current traces and sensitive control signals. The layout was designed according to the EMA-Lab PCB design rules for the DC motor controller project. The board outline was defined in metric units and with full millimeter dimensions. For a two-layer PCB without special mechanical requirements, the specified board thickness is 1.55 mm. The manufacturer

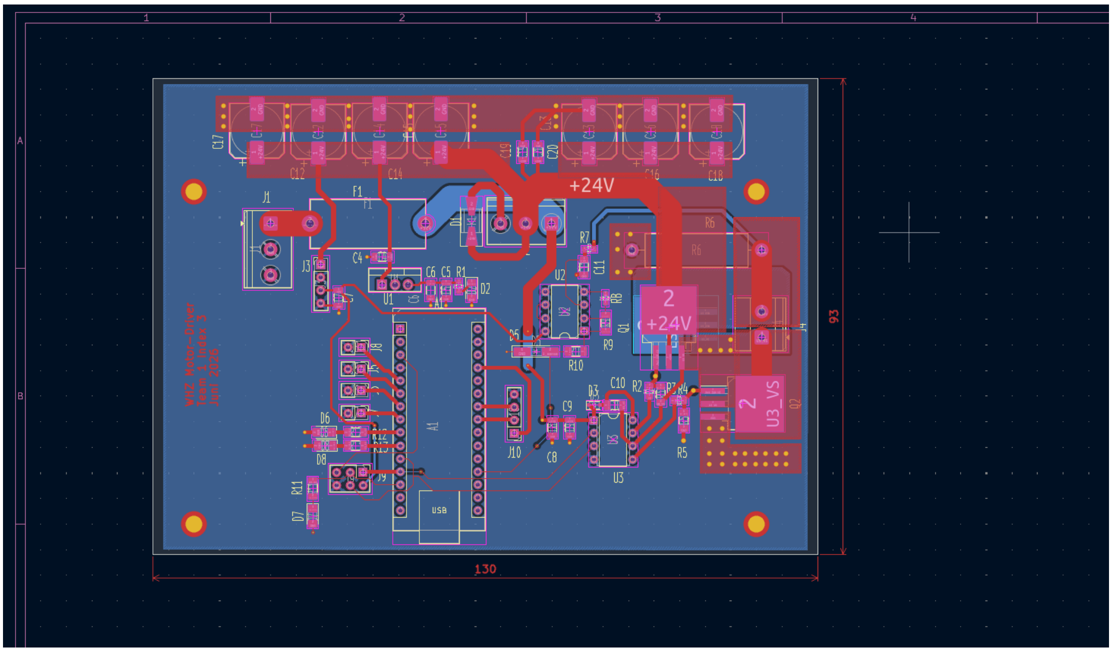


Figure 2.15: PCB layout of the DC motor driver showing the arrangement of the power stage, input capacitors and control electronics.

standard copper thickness is 35 μm after galvanic plating, while a 70 μm copper option may be used if required by the current carrying capability. The minimum trace width and minimum trace spacing specified by the project design rules are both 0.2 mm. Vias require a minimum finished hole diameter of 0.3 mm and a minimum annular ring of 0.25 mm. For the solder mask, a minimum solder mask web width of 0.1 mm and a solder mask clearance of 0.1 mm were considered. The minimum electrical clearance was evaluated according to EN 60664-1:2003. For the voltage levels used in this project, the required minimum clearance was determined as 0.2 mm, including an additional safety margin. Therefore, the general design rule of 0.2 mm minimum spacing was also sufficient with respect to creepage and clearance requirements. Nevertheless, larger distances were preferred in the power stage wherever the available board space allowed it, especially around the 24 V input path, the fuse, the MOSFETs, the freewheel diode and the motor connector. The current carrying capability of the power traces was checked according to IPC-2221 [7]. The required conductor width was calculated from the expected motor current and the permissible temperature rise of the copper trace. The empirical IPC-2221 relation can be written as [7]

$$I = k\Delta T^b A^c, \tag{2.44}$$

where I is the current, ΔT is the allowed temperature rise and A is the trace cross-sectional area. The constants k, b and c depend on whether the trace is located on an external or internal layer. Solving this relation for the required cross-sectional area gives

$$A = \left(\frac{I}{k\Delta T^b}\right)^{\frac{1}{c}}. \tag{2.45}$$

With the copper thickness t_{Cu}, the required trace width follows from

$$w = \frac{A}{t_{\mathrm{Cu}}}. \tag{2.46}$$

For the selected operating point, this calculation resulted in a minimum required conductor width of approximately 0.35 mm. This value is above the manufacturing minimum of 0.2 mm.

However, the actual PCB layout uses substantially wider copper areas for the main 24 V, ground and motor current paths. The IPC-2221 result was therefore used as a lower bound rather than as the final trace width. Particular attention was paid to the arrangement of the input capacitors and the half-bridge power path. The electrolytic capacitors and ceramic decoupling capacitors are placed close to the 24 V supply rail in order to reduce the effective loop impedance and to provide local energy storage for the switching stage [1]. This is important because the motor current is pulsed by the PWM operation and causes current ripple in the supply path. By placing the bulk capacitance close to the power stage, the high-frequency current loop can be shortened and the voltage ripple at the switching node can be reduced. The MOSFETs, freewheel diode and shunt resistor were placed in the right-hand power section of the PCB. This keeps the high-current path compact and avoids routing the motor current through the control section. The gate driver is located close to the MOSFET gates to reduce the gate loop length and to improve switching behavior, in accordance with the layout guidelines for high-side MOSFET drivers [4]. The bootstrap components and gate resistors are also placed near the driver and the MOSFETs, because these components are directly involved in the fast switching process. In contrast, the Arduino Nano and the user interface connectors are placed away from the main power path to reduce coupling from switching currents into the control electronics [5]. The PCB layout also considers thermal aspects. The thermal analysis identified the high-side MOSFET and the freewheel diode as the most critical components. Therefore, these components were connected to larger copper areas to improve heat spreading into the PCB [7]. The wide 24 V and ground copper regions not only reduce ohmic losses, but also act as local heat-spreading structures. Since the board is limited to two copper layers, the available copper area is an important part of the passive thermal management strategy. The PCB design follows a functional separation between the power stage, the voltage regulation and the control electronics. The high-current path is routed with wide copper regions, while the signal traces are kept shorter and separated from the switching node where possible. The minimum manufacturing requirements from the EMA-Lab design rules are fulfilled, and the electrical clearance was checked against EN 60664-1:2003. The IPC-2221 trace width calculation shows that the theoretical minimum conductor width is below the actually used copper areas in the power path. Therefore, the layout provides additional margin with respect to current carrying capability, voltage clearance and thermal spreading.

Chapter 3

Software Design

The software of the motor controller was designed as a state-based embedded control system. Its main tasks are the safe activation of the motor output, the generation of the PWM duty cycle, the measurement and filtering of the motor current, the closed-loop current control and the supervision of critical operating conditions. The implemented software structure is based on four main states: *Idle*, *Start*, *Run* and *Fault*. In the *Idle* state, the PWM output is disabled and the motor is not driven. This state is entered after power-up and after a controlled stop command. The *Start* state is used to perform a soft start, where the duty cycle or current reference is increased gradually instead of applying the full command immediately. After the soft-start phase is completed, the controller enters the *Run* state, where the PI current controller determines the PWM duty cycle during normal operation. The *Fault* state is entered if a critical condition such as an overcurrent or a locked motor condition is detected. In this state, the PWM output is disabled in order to prevent damage to the power electronics, the motor and the supply path. The controller can resume operation only after the fault condition has been removed and the system has returned to a safe operating range. The resulting state machine is shown in Figure 3.1. The state machine separates normal operation from startup and fault handling. This is important because the motor driver must not generate uncontrolled PWM signals during initialization or during a detected fault condition. The soft start is implemented to reduce electrical and mechanical stress during

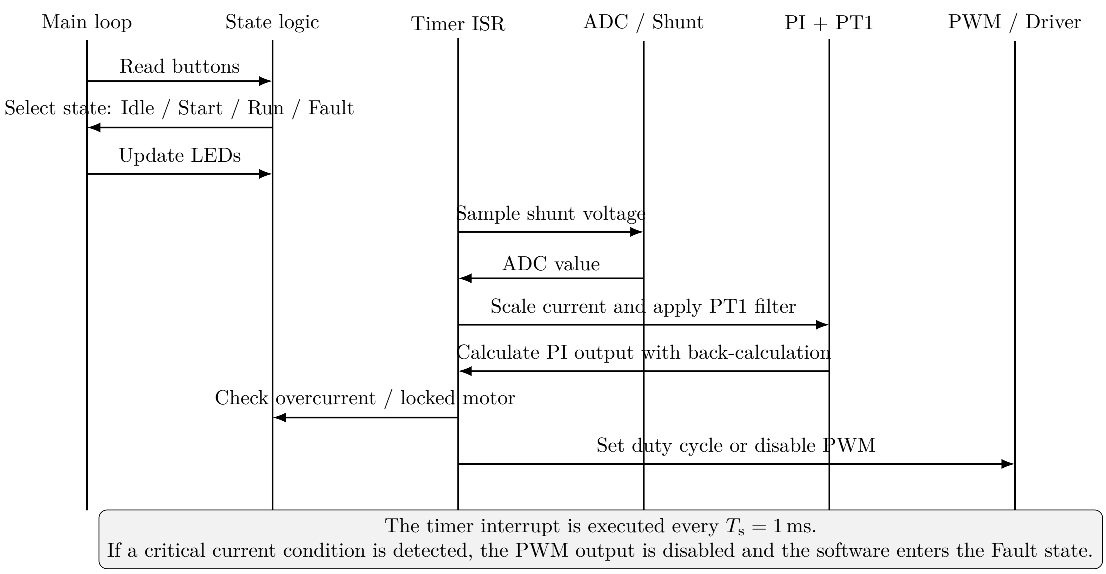


Figure 3.1: Simplified sequence diagram of the embedded motor control software.

motor startup. Without a soft start, the motor current could rise abruptly because the motor initially has no back electromotive force. This may cause a high inrush current and increase the thermal stress of the MOSFETs, the diode, the shunt resistor and the motor winding. Therefore, the controller does not directly apply the final reference value. Instead, the reference is increased step by step until the commanded operating point is reached. With the sampling time T_s, the ramp can be described as

$$r[k] = \min\left(r[k-1] + \Delta r, r_\mathrm{cmd}\right), \tag{3.1}$$

where $r[k]$ is the limited reference value, r_cmd is the commanded reference and Δr is the maximum reference increment per control step. This approach ensures that the PI controller receives a smooth reference input and that the duty cycle does not jump to a high value immediately after enabling the motor. The control algorithm is executed in a timer interrupt service routine with a fixed sampling time of

$$T_\mathrm{s} = 0.001\,\mathrm{s}. \tag{3.2}$$

This corresponds to a control frequency of 1 kHz. The same sampling time is used for the discrete PI controller and for the PT1 filter of the current measurement. Using a fixed interrupt period is important because the discrete controller coefficients are derived for this sampling time. If the control loop were executed with a variable period, the effective controller dynamics would change and the anti-windup behavior would no longer match the design assumptions. The current is measured indirectly by evaluating the voltage across the shunt resistor with the Arduino analog-to-digital converter. The ADC returns a digital value n_ADC, which is converted into an input voltage according to

$$V_\mathrm{ADC}[k] = \frac{n_\mathrm{ADC}[k]}{N_\mathrm{ADC} - 1} V_\mathrm{ref}, \tag{3.3}$$

where N_ADC is the ADC resolution and V_ref is the analog reference voltage. If an amplifier with gain G_shunt is used, the motor current is calculated from the measured voltage by

$$i_\mathrm{meas}[k] = \frac{V_\mathrm{ADC}[k] - V_\mathrm{offset}}{G_\mathrm{shunt} R_\mathrm{shunt}}. \tag{3.4}$$

Here, R_shunt is the shunt resistance and V_offset represents the measurement offset. This offset is required if the measurement circuit does not produce exactly zero volts at zero current or if a biased measurement is used. The calculated current is then filtered before it is used by the controller and the protection logic. The shunt signal is affected by PWM switching noise and ADC quantization noise. Therefore, a first-order low-pass filter is applied to the measured current. The continuous-time transfer function of the PT1 filter is

$$G_\mathrm{f}(s) = \frac{1}{T_\mathrm{f} s + 1}, \tag{3.5}$$

where T_f is the filter time constant. A PT1 filter is suitable for this application because it reduces high-frequency measurement noise while preserving the slower current dynamics required by the controller. The filter was discretized using the Tustin approximation

$$s \approx \frac{2}{T_\mathrm{s}} \frac{z-1}{z+1}. \tag{3.6}$$

Substituting this approximation into the PT1 transfer function gives the discrete filter relation

$$y[k] = a_\mathrm{f} y[k-1] + b_\mathrm{f}\left(x[k] + x[k-1]\right), \tag{3.7}$$

with

$$a_\mathrm{f} = \frac{2T_\mathrm{f} - T_\mathrm{s}}{2T_\mathrm{f} + T_\mathrm{s}}, \tag{3.8}$$

and

$$b_{\mathrm{f}} = \frac{T_{\mathrm{s}}}{2T_{\mathrm{f}} + T_{\mathrm{s}}}. \tag{3.9}$$

In this equation, $x[k]$ is the unfiltered current value and $y[k]$ is the filtered current value. The filter is implemented in incremental form by storing the previous input and output values between interrupt calls. For current control, a PI controller was selected instead of a PID controller. A proportional term reacts directly to the current error, while the integral term removes steady-state error. This is useful because the motor, the MOSFETs and the wiring introduce voltage drops that may otherwise lead to a persistent current error. A derivative term was not used because the derivative action would amplify the measurement noise of the shunt signal. Since the current measurement is already affected by PWM ripple and ADC noise, a PID controller would increase the sensitivity of the duty cycle to measurement disturbances. Therefore, the PI controller represents a more robust choice for this task. The continuous-time PI controller is given by

$$G_{\mathrm{PI}}(s) = K_{\mathrm{p}} + \frac{K_{\mathrm{i}}}{s}, \tag{3.10}$$

where K_{p} is the proportional gain and K_{i} is the integral gain. Using the Tustin approximation, the controller can be implemented as an incremental discrete-time controller:

$$u_{\mathrm{PI}}[k] = u_{\mathrm{PI}}[k-1] + b_0 e[k] + b_1 e[k-1], \tag{3.11}$$

with

$$b_0 = K_{\mathrm{p}} + \frac{K_{\mathrm{i}} T_{\mathrm{s}}}{2}, \tag{3.12}$$

and

$$b_1 = -K_{\mathrm{p}} + \frac{K_{\mathrm{i}} T_{\mathrm{s}}}{2}. \tag{3.13}$$

The control error is defined as

$$e[k] = i_{\mathrm{ref}}[k] - i_{\mathrm{filt}}[k], \tag{3.14}$$

where $i_{\mathrm{ref}}[k]$ is the current reference after the soft-start ramp and $i_{\mathrm{filt}}[k]$ is the filtered current measurement. The controller output is limited to the valid PWM duty-cycle range. Therefore, the unsaturated duty cycle $u_{\mathrm{PI}}[k]$ is converted into the applied duty cycle

$$D[k] = \mathrm{sat}\left(u_{\mathrm{PI}}[k]\right), \tag{3.15}$$

with

$$0 \le D[k] \le D_{\mathrm{max}}. \tag{3.16}$$

Since the duty cycle is saturated, anti-windup is required to prevent the integral part of the PI controller from accumulating an unrealistically large value during saturation. For this purpose, back-calculation was used. The difference between the saturated controller output and the unsaturated controller output is fed back into the integrator path:

$$e_{\mathrm{aw}}[k] = D[k] - u_{\mathrm{PI}}[k]. \tag{3.17}$$

The anti-windup correction reduces the internal controller state when the actuator is saturated and therefore improves the recovery behavior once the controller leaves saturation. The structure of the current control loop is shown in Figure 3.2. The ADC measurement is converted into a current value, filtered by the PT1 filter and compared with the current reference. The PI controller with back-calculation then computes the PWM duty cycle. The resulting duty cycle is applied to the motor driver, while the same measured current is also used by the protection logic. The protection logic is executed together with the control loop. The most important mandatory fault cases are a locked motor and overcurrent conditions.

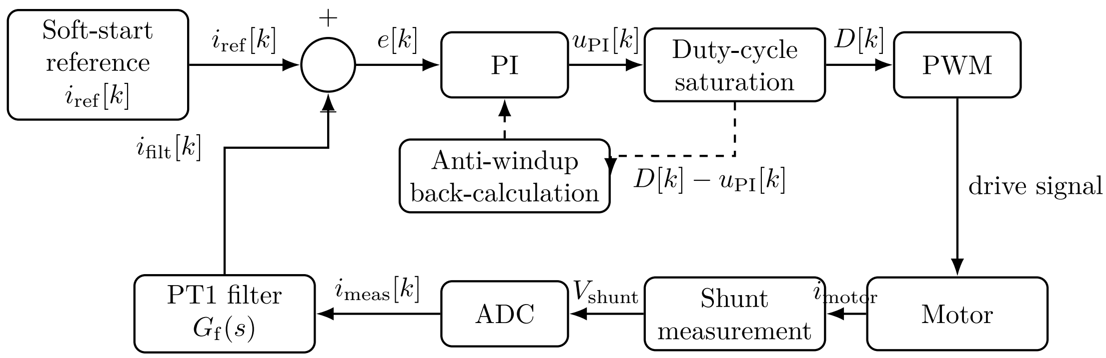


Figure 3.2: Current control loop with shunt measurement, PT1 filtering, PI control, duty-cycle saturation and back-calculation anti-windup.

A locked motor can lead to a high current because the motor does not generate sufficient back electromotive force. Therefore, the software monitors the measured current and an accumulated I^2R-related load value. The instantaneous ohmic loss can be approximated by

$$P_R[k] = i_{\mathrm{filt}}^2[k]R, \tag{3.18}$$

where R represents the relevant resistance of the monitored current path. A simple accumulated stress value can be calculated as

$$E_R[k] = E_R[k-1] + P_R[k]T_{\mathrm{s}}. \tag{3.19}$$

This value increases when a high current is present for a longer time. In practical implementation, this supervision may be combined with a direct current threshold to react quickly to severe overcurrent events:

$$i_{\mathrm{filt}}[k] > i_{\mathrm{max}}. \tag{3.20}$$

If the threshold is exceeded or the accumulated load becomes too high, the controller enters the *Fault* state and disables the PWM output. This behavior addresses the requirement that a locked motor or overcurrent fault must not damage the system or its components. After the blockage has been removed and the measured current has returned to a safe range, the software can leave the *Fault* state through a reset or restart condition. The optional fault cases, such as overtemperature, overvoltage and undervoltage, were treated separately from the mandatory current-related protection. Since no dedicated temperature sensor or voltage monitoring channel was implemented in the basic software version, these functions were not part of the active fault handling. Their implementation would only be required if the risk assessment shows that the circuit can be damaged by these conditions without additional protection. In the present implementation, the main protective mechanism is therefore the current-based supervision of overcurrent and locked-motor conditions. The buttons and LEDs are integrated into the cyclic software execution. The buttons are used to enable or disable the motor output and to adjust the command value, for example by increasing or decreasing the reference. In order to avoid false triggering due to mechanical contact bounce, button states should be evaluated with a simple debounce mechanism or edge detection. The LEDs are used as optical indicators for the current software state and for detected faults. For example, an LED can indicate whether the controller is in *Idle*, *Run* or *Fault*. Additional indicators for overcurrent, speed-up or speed-down commands are optional according to the project requirements. The sequence of one software cycle is shown in Figure 3.1. The main loop handles non-time-critical tasks such as button evaluation, state transitions and LED updates. The timer interrupt provides the deterministic control timing. Inside the interrupt, the current measurement is sampled, converted, filtered and used by the

PI controller. The protection logic is evaluated before the PWM duty cycle is applied. If a fault is detected, the PWM output is disabled immediately and the state machine enters the *Fault* state. The software design therefore combines deterministic interrupt-based control with a state machine for safe operation. The 1 ms interrupt period defines the sampling time for the PT1 current filter and the discrete PI controller. The soft-start function prevents abrupt current commands during startup, while the PI controller with back-calculation generates a saturated PWM duty cycle without excessive integrator windup. The shunt-based current measurement is used both for closed-loop control and for protection against locked-motor and overcurrent conditions. Optional functions such as overtemperature, overvoltage and undervoltage detection can be added if additional sensors or measurement channels are included in the hardware. “‘latex

Chapter 4

Conclusion and Outlook

This chapter summarizes the main results and lessons learned during the development of the 24 V brushed DC motor driver. The project combined circuit design, PCB layout, embedded software development and basic verification of the selected design decisions. Therefore, the conclusion does not only evaluate the final technical solution, but also reflects the practical development process. The first section summarizes the achieved results, the gained experience and the remaining weaknesses of the current prototype. The second section gives an outlook on possible improvements and describes how the knowledge gained in this project can be extended in future work.

4.1 Conclusion and Weaknesses

The project provided practical experience in the complete development process of an embedded power electronic system. Starting from the system requirements, the circuit was divided into functional blocks such as power supply, gate driver, half-bridge power stage, current measurement, user interface and microcontroller control. During the circuit design phase, the team gained a better understanding of the interaction between power electronics and embedded control. Important design aspects were the selection of suitable MOSFETs, the use of a gate driver for the half-bridge, the dimensioning of the input capacitance, the placement of ceramic decoupling capacitors, the use of a shunt resistor for current measurement and the thermal loading of the main power components. The project also showed that even a comparatively simple brushed DC motor driver requires careful consideration of current paths, switching behavior, voltage levels and protection mechanisms. A major part of the learning process was related to the used development tools. Proteus was used for schematic design and circuit-related work, while PlatformIO was used for embedded software development. Compared to the standard Arduino IDE, PlatformIO provides a more structured development environment, especially for larger projects with multiple source files, header files and reusable software modules. This was useful for implementing the control software in C++ for the Arduino Nano. The software development improved the team's understanding of embedded programming concepts such as timer-based execution, interrupt service routines, fixed sampling times, state-based software logic and hardware-related input and output handling. The implementation of the soft-start function, the PI controller with back-calculation, the PT1 filter for the shunt measurement and the current-based fault supervision required a closer connection between control theory and practical microcontroller programming. The software was structured around the logical states *Idle*, *Start*, *Run* and *Fault*. This structure helped to separate safe inactive behavior, controlled startup, normal operation and fault handling. The soft-start function reduces the risk of abrupt current peaks during motor startup by increasing the reference gradually. The PI controller was selected because it can compensate steady-state current errors while avoiding the measurement-noise amplification that would be introduced by a derivative term. Since the controller output is limited by the physically possible PWM duty-cycle range, back-calculation anti-windup was included to prevent excessive integrator windup during saturation. The shunt measurement was filtered with a discrete PT1 filter, because the raw current signal is expected to contain PWM ripple, ADC quantization effects and switching noise. The timer interrupt with a sampling time of $T_\mathrm{s} = 0.001\,\mathrm{s}$ provides a deterministic execution period for the filter and the controller. The PCB design phase showed that the physical implementation of a motor driver is more complex than the schematic alone suggests. The design had to consider minimum trace spacing, creepage and clearance, current carrying capability, component placement, connector positions and thermal spreading. The power path had to be routed with wide copper areas, while the control signals had to be kept away from the switching and high-current regions as far as possible. The placement of the electrolytic and ceramic capacitors near the supply and power stage was important to reduce the effective loop impedance and to support the switching current demand of the half-bridge. The PCB layout also showed the practical relevance of the thermal analysis, because the MOSFETs, the freewheel diode and the shunt resistor are not only electrical components, but also heat sources that require sufficient copper area and careful placement. At the same time, the project still contains several weaknesses and simplifications. The PI controller has not yet been fully parameterized based on measured motor data. For a reliable controller design, measurements with the real motor are still required. These measurements should include the current response to different PWM duty cycles, the behavior during startup and the current behavior under load. Without these data, the controller parameters can only be chosen based on assumptions and practical tuning. The same limitation applies to the PT1 filter. A suitable cutoff frequency should be selected based on measured shunt signals, because the filter must reduce PWM-related noise without making the current feedback too slow for the controller and the protection logic. Another simplification is the current-based protection strategy. The software considers overcurrent and locked-motor behavior by monitoring the measured current and an accumulated I^2R-related load value. This is a reasonable first approach, but the thresholds still require experimental validation. A blocked motor must not damage the system, and overcurrent faults must be detected fast enough to protect the MOSFETs, the diode, the shunt resistor and the motor. Therefore, the current limits and the fault recovery behavior still need to be tested with the real hardware. Optional protection functions such as overtemperature, overvoltage and undervoltage were not fully implemented in the basic software version. This is acceptable if the damage risk is covered by the existing hardware design and laboratory operation, but these functions would be necessary for a more robust standalone system. The thermal analysis was also simplified. The component losses and temperatures were estimated using analytical equations and lumped thermal models. This approach is useful for identifying critical components and comparing operating conditions, but it does not replace measurements on the assembled PCB. The actual temperature depends on the solder joints, copper area, airflow, component tolerances and heat transfer through the board. Therefore, the high-side MOSFET, the freewheel diode and the shunt resistor should be checked experimentally during operation. The PCB design itself also took longer than expected. The routing of the power stage, the placement of the capacitors and the fulfillment of the design rules required more iteration than initially planned. Without supervision and feedback from the laboratory staff, the PCB design would likely not have reached a manufacturable state in the available time. This was an important practical lesson, because it showed that PCB design is not only a drawing task, but a system-level engineering task that requires experience, design rules and review.

4.2 Outlook

The next development step should be the experimental validation of the assembled motor driver. First, the power supply, voltage regulation and gate driver signals should be tested without the motor connected. After that, the PWM output and the half-bridge behavior should be checked with a limited current supply and a safe load. Only then should tests with the real 24 V motor be performed. During these tests, the motor current, shunt voltage, PWM duty cycle and component temperatures should be measured. These data can then be used to tune the PI controller and to determine a suitable cutoff frequency for the PT1 filter. A useful improvement would be a systematic controller tuning procedure. Instead of choosing the PI parameters manually, the motor and power stage could be identified from measurements. A step response of the current could be recorded and used to estimate a simple plant model. Based on this model, the PI controller parameters could be selected more systematically. The same measurements could be used to compare different filter time constants and to evaluate the trade-off between noise suppression and control delay. This would make the software design more reproducible and would improve the connection between control theory and practical implementation. The protection concept could also be extended. A hard overcurrent threshold should be combined with the existing I^2R-based supervision, so that severe current peaks are detected immediately while longer overload conditions are handled by the accumulated thermal load estimate. Additional voltage monitoring could be added to detect undervoltage and overvoltage conditions. If the thermal risk of the power stage is considered critical, temperature sensors near the MOSFETs or the diode could be included. These additions would make the system more robust and would allow the software to react to a wider range of fault cases. On the hardware side, the PCB could be improved after the first test results are available. The copper areas around the MOSFETs, diode and shunt resistor could be enlarged if the measured temperatures are too high. Thermal vias or a different component placement could improve heat spreading. The routing of the power loop could also be optimized further to reduce parasitic inductance and switching noise. If the current ripple or voltage ripple is higher than expected, the input capacitance and the ceramic decoupling network should be reviewed. A future design iteration could also include more test points, because accessible measurement points make debugging and validation much easier. The knowledge gained in this project can be reused and extended in future embedded power electronics projects. The experience with PlatformIO and C++ can be applied to more structured firmware projects, where drivers, filters, controllers and state machines are separated into reusable modules. The experience with Proteus, PCB design rules and practical routing can be used for future hardware projects with higher complexity. The control concepts used here, especially discrete-time filtering, PI control, PWM generation and anti-windup, are not limited to brushed DC motors. They are also relevant for other electric drive systems, power converters and mechatronic systems. Future work could therefore extend the system towards bidirectional motor operation, speed control, more advanced current limiting or a different microcontroller platform with better ADC performance and more precise timer peripherals.

Appendix A

Appendix

A.1 Complete Bill of Materials

Table A.1 lists all components, connection materials and mechanical hardware used in the 24 V / 250 W brushed DC motor driver. The BOM corresponds to the frozen revision `BOM_ElectricDriveSystems_1_1.xlsx` submitted to Prof. Zitzelsberger.

Table A.1: Complete Bill of Materials.

#	Ref.	Qty	Value / Description	Spec	Unit €
A — PCB Electronic Components					
A.1 — Power Stage					
1	Q1, Q2	2	IRFZ44NSPBF MOSFET N-Ch	D^2Pak, 55 V 49 A	0.91
2	D4	1	MBRS2045CT Schottky diode	D^2Pak, 20 A 45 V	0.90
3	F1	1	LITT 0287015 Fuse	KFZ ATO, 15 A	0.87
4	D1	1	P4SMAJ33CA TVS diode	Bidirect., 33 V, 400 W, SMA	0.10
5	D3	2	BAS 16J	SOD-323F, 100 V, 0.25 A	0.03
6	D5	1	BZV 55-C5V1 Zener	5.1 V, 0.5 W, SOD-80	0.02
7	—	2	FK 244 13 D2PAK heatsink	13 mm Cu, 22.8 K/W	0.99
A.2 — Voltage Regulation					
8	U1	1	L7812CV linear regulator	TO-220, 12 V	0.23
9	J3	1	LM2596S buck module	24 V → 5 V, 2.5 W	2.99
A.3 — Control & Driver					
10	A1	1	Arduino Nano v3.x (ATmega328P)	16 MHz, 5 V module	—
11	U3	1	IR2104 gate driver	DIP-8	1.30
12	U2	1	LM358 dual op-amp	DIP-8	0.25
A.4 — Capacitors					
13	C1	8	Hybrid polymer elko 100 μF	SMD, 50 V, 145 °C	1.70
14	C3,C6,C7,C9,C11	10	100 nF X7R MLCC	1206, 50 V	0.07
15	C4	5	330 nF X7R MLCC	1206, 50 V	0.07
16	C5	10	10 μF X7R MLCC	1206, 50 V	0.46
17	C2,C8,C10	10	1 μF X7R MLCC	1206, 100 V	0.14

Continued on next page

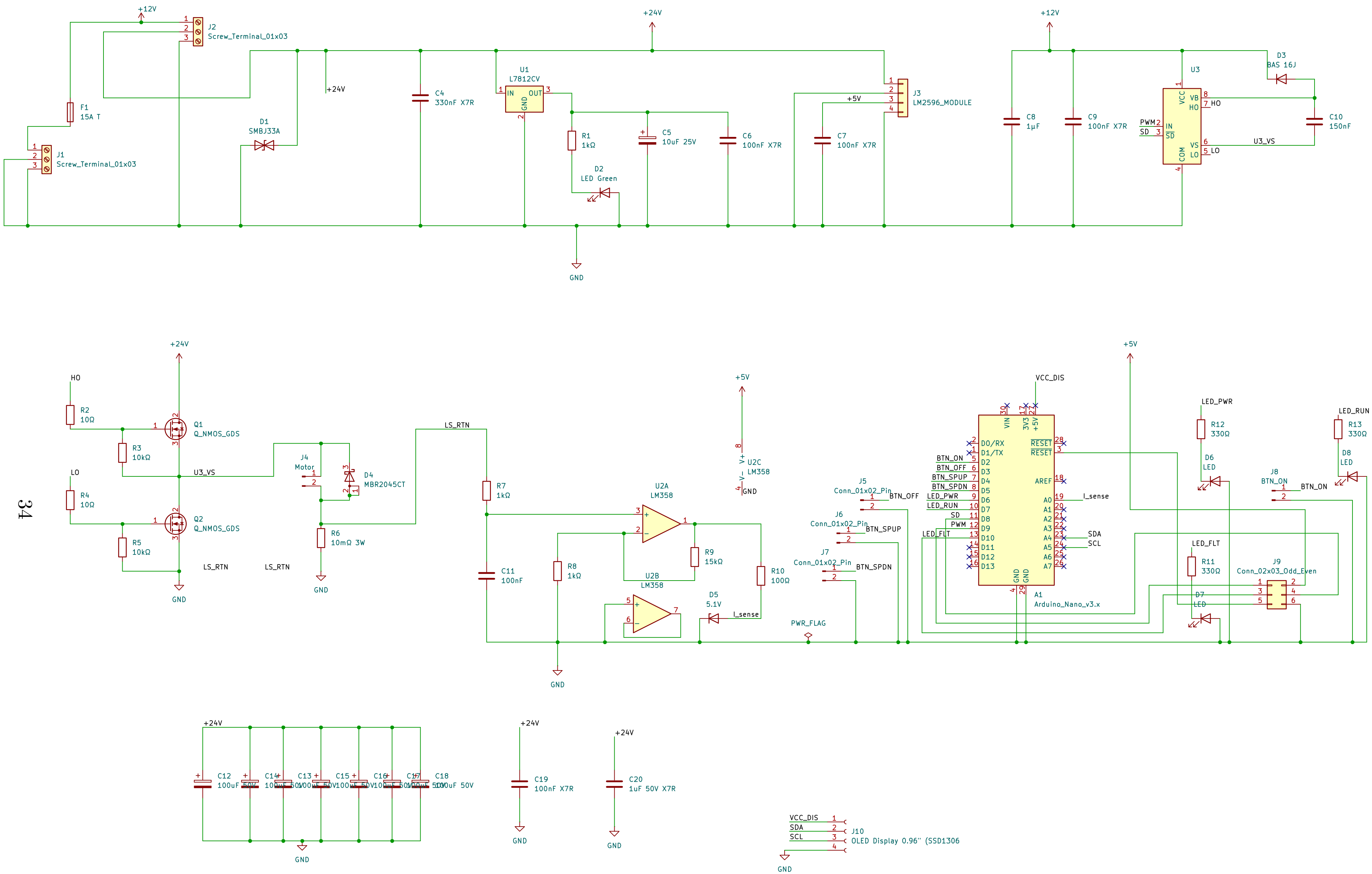
#	Ref.	Qty	Value / Description	Spec	Unit €
A.5 — Resistors					
18	R1,R7,R8	3	93.1 kΩ 1%	0603, 100 mW	0.02
19	R2,R4	2	10 Ω 5%	0805, 125 mW	0.02
20	R3,R5	2	10 kΩ 1%	1206, 250 mW	0.02
21	R6	1	10 mΩ shunt	Axial, 4 W	0.75
22	R9	1	15 kΩ 1%	1206, 250 mW	0.02
23	R10	1	100 Ω 1%	1206, 250 mW	0.09
24	R11,R12,R13	3	330 Ω 1%	1206, 250 mW	0.02
A.6 — LEDs & Indicators					
25	D2	2	Super yellow-green LED	1206, 29 mcd	0.05
26	—	1	Red LED	1206, 180 mcd	0.07
27	D6,D7,D8	1	Yellow LED	1206, 150 mcd	0.09
A.7 — PCB Connectors					
28	J1	1	Screw terminal 2-pole	5.08 mm, 24 V input	0.33
29	J2	1	Screw terminal 2-pole	5.08 mm, main power	0.33
30	J4	1	Screw terminal 2-pole	5.08 mm, motor output	0.33
31	J5,J6,J7	3	Pin header 1×02	2.54 mm	0.06
32	J8	1	Pin header 1×02	2.54 mm, ON button	0.06
33	J9	1	Pin header 2×03	2.54 mm, ISP	0.16
34	—	2	Socket header 1×16	2.54 mm, Arduino	0.99
35	—	1	Fuse holder 5×20 mm	Horizontal mount	0.27
B — Connection Materials					
B.1 — Cables & Wires					
36	W1	1	12 AWG silicone wire RED 5 m	600 V, stranded	11.99
37	W2	1	12 AWG silicone wire BLACK 5 m	600 V, stranded	11.99
38	—	1	OLED display 0.96" SSD1306	I^2C	6.99
39	W3	1	22 AWG hookup wire set (6 col.)	10 m per colour	14.89
40	W4	1	18 AWG wire RED+BLACK 3 m ea.	Stranded	3.10
B.2 — Off-Board Connectors					
41	CON1	1	XT60 panel-mount pair	60 A rated	2.99
42	CON2	2	4 mm banana socket	32 A, panel-mount	1.85
43	CON3	1	GX16-2 aviation connector pair	2-pin, 15 A	2.99
B.3 — Terminals & Crimps					
44	T1	4	Screw terminal block (spare)	5.08 mm, 15 A	0.33
45	T2	1	Crimp ferrule kit 0.5–6 mm^2	∼100 pcs	4.90
46	T3	1	Heat-shrink assortment	100 pcs, 2:1	3.50
47	T4	20	Ring terminal M3/M4	4 mm^2	0.20
B.4 — Cable Management					
48	CM1	1	Cable tie pack 100+200 mm	100 pcs	0.50
49	CM2	20	Self-adhesive cable clips	Wood screw mount	0.37

Continued on next page

#	Ref.	Qty	Value / Description	Spec	Unit €
50	CM3	1	Spiral wrap 6 mm × 2 m	PE, black	1.45
C — Mechanical & Mounting Hardware					
C.1 — Board & Fasteners					
51	BD1	1	Plywood board 400×300×12 mm	Birch, sanded	14.76
52	BD2	1	M3 brass standoff set	F-F 10 mm + screws	13.20
53	BD3	10	M4×16 wood screws + washers	—	1.99
54	BD4	1	M3 spacer set 680 pcs	Hex standoffs, brass	14.59
55	BD5	4	Rubber feet (self-adhesive)	⌀15 mm, 5 mm height	1.90
C.2 — Switches & Buttons					
56	SW1	1	Toggle switch 16 A 250 V	SPST, panel-mount	2.60
57	SW2–SW5	4	Push button 12 mm momentary	PBS-33B, 4 colours	2.29
C.3 — Thermal Management					
58	HS1	2	Heatsink for IRFZ44N (TO-220)	SK 129, 38.1 mm	2.70
59	HS2	1	Heatsink for L7812 (TO-220)	SK 104, 38.1 mm	1.99
60	HS3	1	Thermal grease Arctic MX-4 4 g	8.5 W/mK	5.49
C.4 — LED Panel Holders					
61	LH1	4	5 mm LED panel mount holder	Chrome bezel, snap-in	7.55
D — DC Motor					
62	M1	1	MY1016 24 V 250 W motor	3000 RPM, 13.7 A	—
63	M2	1	Motor L-bracket	Steel, 2× M6	11.56
64	M3	1	100 nF disc cap (brush suppress.)	100 V ceramic	0.04
E — PCB Fabrication					
65	PCB1	1	PCB fab. 5 pcs (2-layer)	1.6 mm FR4, HASL-LF, 2 oz Cu	—

A.2 KiCad Schematic

The complete circuit schematic of the 24 V / 250 W motor driver is shown on the following page. The schematic was created in KiCad and exported as a single-sheet landscape drawing.



ELT10140 - Team 1

24V/250W DC Motor Driver

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